

# Contact Delassus operators as weighted Laplacians on the contact line graph

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2026-09-20

## Source key for the inline citations

Every number below carries the file and section it was measured in. The keys resolve as:

| key      | file (relative to the measurement archive root)                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08       | 08_math_transfer_into_solver.md                                                                                                                                          |
| h11      | build/h11/RESULTS.md                                                                                                                                                     |
| h13      | build/h13/RESULTS.md                                                                                                                                                     |
| h15      | build/h15/RESULTS.md                                                                                                                                                     |
| h1 / h1b | build/h1/RESULTS.md (§H1b is the GPU port inside the same file)                                                                                                          |
| h23      | build/h23/RESULTS.md                                                                                                                                                     |
| U18      | build/U18/RESULTS.md                                                                                                                                                     |
| U20      | build/U20/RESULTS.md (uncontended Modal L4 timings)                                                                                                                      |
| U21      | build/U21/RESULTS.md                                                                                                                                                     |
| U22      | build/U22/RESULTS.md                                                                                                                                                     |
| U33      | build/U33/RESULTS.md                                                                                                                                                     |
| U55      | build/U55/RESULTS.md                                                                                                                                                     |
| U83      | build/U83/RESULTS.md                                                                                                                                                     |
| U87      | build/U87/RESULTS.md                                                                                                                                                     |
| U89      | build/U89/RESULTS.md (data file<br>build/U89/p1_figures_data.csv)                                                                                                        |
| U95      | build/U95/RESULTS.md                                                                                                                                                     |
| A63      | build/A63/RESULTS.md (audit of U67 and U56)                                                                                                                              |
| A68      | build/A68/RESULTS.md (audit of U95)                                                                                                                                      |
| A70      | build/A70/RESULTS.md (audit of U89; restrictions in<br>build/A70/field_restrictions.csv)                                                                                 |
| U126     | build/U126/RESULTS.md                                                                                                                                                    |
| U131     | build/U131/RESULTS.md (per-direction data<br>build/U131/directions.csv)                                                                                                  |
| U132     | build/U132/RESULTS.md (figure data<br>build/U132/p1_figures_data_v2.csv, per-cell status<br>build/U132/field_provenance.csv, diff against U89<br>build/U132/changes.csv) |
| A73      | build/A73/RESULTS.md (audit of U131)                                                                                                                                     |
| U134     | build/U134/RESULTS.md (step against ramp; per-direction data<br>build/U134/analysis.json)                                                                                |
| A74      | build/A74/RESULTS.md (audit of U134)                                                                                                                                     |
| A77      | build/A77/RESULTS.md (audit of h13 patch lumping)                                                                                                                        |
| A78      | build/A78/RESULTS.md (audit of U18 router)                                                                                                                               |
| A79      | build/A79/RESULTS.md (audit of h11)                                                                                                                                      |
| A80      | build/A80/RESULTS.md (audit of h15)                                                                                                                                      |
| A81      | build/A81/RESULTS.md (audit of U21 and U22)                                                                                                                              |
| A82      | build/A82/RESULTS.md (audit of 08 §1)                                                                                                                                    |

| key     | file (relative to the measurement archive root)                                                                   |
|---------|-------------------------------------------------------------------------------------------------------------------|
| U135    | build/U135/RESULTS.md (25-point $\rho$ sweep over the 21 scenes of table 6; table data build/U135/tabell6_v2.csv) |
| digests | build/digests/RESULTS.md                                                                                          |
| HUNT    | HUNT.md, section “Mätt”                                                                                           |

build/U67/p1\_figures\_data.csv is withdrawn as a data source: 26 of 36 checked fields — 9 C1 labels, 14 degrees, 3 contact counts — disagree with the run code [U89 §1; A63 §“U67: nya processer och figurfil”]. The figures of this draft read build/U132/p1\_figures\_data\_v2.csv, which re-measures 213 cells over the same 36 rows from the run code and leaves 41 cells empty [U132 §2]. Table 6 of §6.6 reads build/U135/tabell6\_v2.csv: 25  $\rho$  values on each of 21 scenes, with the same operator used for the spectrum, the ADMM x-step, the de Saxcé shift and the residual [U135 §§Operator och lösare, Tabell]. The oracle column of build/U89/p1\_figures\_data.csv remains withdrawn because on 10 of its 21 rows it equals the prediction against which it was scored [A70 §7].

build/math/RESULTS.md does not exist — the build/math/ lane wrote its results into 08, and the scripts (t1\_delassus\_laplacian.py ... t7\_oed\_mu\_identifiability.py) sit next to them. Citations that the brief expected to read [math §...] therefore read [08 §1]. [MISSING: build/math/RESULTS.md]

## Abstract

Contact solvers for rigid multibody systems slow down as the mass ratio across a contact chain grows, and the operator that carries that slowdown — the Delassus operator  $G = J M^{-1} J^T$  — is used in [16], [17] and [21] through its assembled entries, its diagonal and its extreme eigenvalues. We show that  $G$  is a weighted Laplacian on the line graph of the contact graph: nodes are contacts, edges are shared bodies, and the conductance of an edge is the shared body’s inverse (spatial) inertia. The identity is exact for one contact per body pair on translation-only bodies with normals within  $90^\circ$  [08 §1], and it becomes exact as a *block* Laplacian — rotation and normal angle absorbed into a  $6 \times 6$  conductance — as soon as each face-face patch is lumped into one node carrying a 6-dimensional wrench; agreement is  $9.6e-17$  relative, off-diagonal blocks are negative semidefinite, block row sums equal the graph’s Dirichlet residue [h13 §1B]. Friction rows do not break the structure ( $G = L \otimes I_3$  exactly for point masses with a common contact frame, i.e. parallel normals [08 §1; Prop. C.5]; for differing frames  $G$  is a connection Laplacian  $L_{\{cc'\} E_c E_{\{c'\}^T}$ ); what breaks it is body degree  $\geq 3$ , where three incidence signs cannot be pairwise opposite and the assembly becomes a *signed* block Laplacian [h13 §1E]. Three consequences are measured. The non-zero spectrum equals that of the vertex pencil ( $L_v, M$ ) to  $1e-15$  [h13 §1G], so the mass-ratio bottleneck is read off the graph in  $\leq 204$  matvecs without forming  $G$ , at  $\lambda_{\min}^+$  relative error  $\leq 1.8e-9$  and  $\lambda_{\max}$  relative error  $\leq 2.1e-7$  over the five scenes measured [h15 §1A; h1 §H1b.1].  $\rho = \rho^*/10$  turns 18 225 PGS sweeps into 216 ADMM iterations on a 1000:1 column, while  $\rho = \lambda_{\min}^+$  gives 213 iterations in that h11 scene [h11 §1A, §1C; A79 §1]. The graph estimate  $\rho = \lambda_{\min}^+$  turns 9 215 into 29 on the set-binding variant [h15 §2], 14 630 into 140 (patch) or 8 (point) on an 8-box tower [h13 §2; h11 §1A], and 1 512 into 9 on an asymmetric tower [h15 §2]. Laplacian preconditioning is scene-dependent: on the h11-b chain, Laplacian/Jacobi/no preconditioner take 1/3/3 median PCG iterations and 2.86/3.84/3.08 ms for  $5 \times 20$  solves, whereas on pack50 they take 62/35/33 iterations and 2.72/0.21/0.17 s [A79 §2]. Patch lumping makes each patch block of  $J$  full row rank, so contact derivatives exist without regularisation (cond  $4.9e16$  / singular  $\rightarrow 72.6$ , analytic  $dw/d\mu$  against central FD  $1.9e-10$ ) [h13 §3]. On chains the x-update is block tridiagonal and the graph solve reproduces the dense Cholesky ADMM iteration for iteration to  $1.7e-16$  [h13 §2D]. We also report what the spectrum is not:  $\lambda_{\min}^+$  is constant to four digits while a tower is tilted to its toppling angle, and it is the certificate margin, not the spectrum, whose zero crossings agree with the ZMP and Coulomb

margins in the stated K8 sweeps. A monotone one-dimensional sweep gives Spearman  $r = 1$  by construction; 40 randomized towers have maximum pointwise margin discrepancy 0.97 [U22 §2, §5; A81]. Spectral  $\rho$  prediction hits 5 of 21 scenes within  $2\times$  in measured sweeps; the rule is not used as the router's primary choice [U135 §Klass och träffdefinition].

## 1. Introduction

The velocity-level contact problem of a rigid multibody step is a cone complementarity problem in the contact impulses  $\lambda$ :

$$u = G \lambda + b, \quad G = J M^{-1} J^T, \quad K_\mu \ni \lambda \perp u + \Gamma(u) \in K_\mu^*$$

with  $\Gamma$  the de Saxcé shift that makes the exact Coulomb cone, rather than its convex relaxation, the object being solved. The operator  $G$  — the Delassus operator — is what every first-order method iterates against, and its conditioning is what sets the iteration count.

**The gap.** Mass-ratio stiffness is the failure mode measured here on projected Gauss–Seidel, and it is treated numerically rather than structurally. Measured here on a 3-box column with masses 1000/1/1000 and the cone binding: plain PGS needs 18 225 sweeps to a natural-map residual of  $1e-8$ , block-Jacobi preconditioning does not converge in 30 000, and a sketched inverse at  $k = 16/64/256$  columns with three seeds does not converge at any of  $\omega \in \{1.0, 0.5, 0.25\}$  [08 §1E; h11 §1A]. An 8-box tower with rotation needs 14 630 sweeps [h13 §2]; an asymmetric 1/1000 tower converges under none of five methods [08 §1F]. The remedies in circulation are compliance (soft contacts), substepping, and tuned relaxation — measured here as well: soft-step collapses in all nine configurations tried at 1000:1, with rest overlap linear in the supported mass ( $\times 9.2$  per decade), so a 0.1 mm target at 1000:1 needs  $\approx 10^4$  Hz [HUNT, B3]; substepping at a fixed iteration budget buys  $2.3\times$  less penetration and leaves the residual unchanged [HUNT, B2].

What is missing is a *structural* reading of  $G$ . The mass ratio is a property of the contact graph. The sources cited here treat  $G$  through its assembled entries; [16], [17] and [21] build preconditioners and step rules from  $G$ 's norms and spectrum rather than from the incidence structure. The candidate identity — that  $G$  is a graph Laplacian — is folklore for the frictionless point-mass case and is usually left there, because three objections apply immediately to any real scene: bodies rotate, a face-face contact carries four corner points rather than one, and friction triples the row count.

This paper states the identity precisely, measures where each objection bites, and shows what each costs: friction is harmless on point masses with a common frame ( $G = L \otimes I_3$ ), rotation and multiplicity are removed by patch lumping, which is an equality of cones on the normal + moment part (Theorem C.9) and a change of contact law on the tangential part, measured at 0.3232 relative wrench error on the spin counterexample [U89 §2]; and what is left — body degree  $\geq 3$  — is a structural limit that turns the Laplacian into a signed Laplacian. It then reports what the structure buys in a solver, and, symmetrically, one thing it was expected to buy and does not. Spectral  $\rho$  prediction hits 5 of 21 scenes within  $2\times$  in measured sweeps; the rule is not used as the router's primary choice [U135 §Klass och träffdefinition].

## 2. Theorem: the contact Delassus is a weighted (block) Laplacian on the line graph of the contact graph

### 2.1 Setup

Let the contact graph have bodies as vertices and contacts as edges. Let  $B$  be the signed contact-body incidence matrix, one row per contact, with the static ground column dropped. Write  $w_i$  for the conductance of body  $i$ .

**Statement (scalar form).** For normal-only rows on translation-only bodies with parallel normals,

$$G = B W B^T, \quad W = M^{-1} = \text{diag}(1/m_i)$$

i.e.  $G$  is a **grounded weighted Laplacian on the line graph** of the contact graph: nodes are contacts, edges are shared bodies, the conductance of edge  $(c, c')$  is the shared body's inverse mass  $1/m_i$ , and a body touched by exactly one contact contributes a Dirichlet (ground) term to that contact's diagonal [08 §1].

**Statement (block form, patch-lumped).** Lump every face-face patch into one node carrying the 6-dimensional wrench  $w = (F_n, M_{t1}, M_{t2}, F_{t1}, F_{t2}, M_n)$  at the patch centroid. Then

$$L[c, c'] = \sum_i \sigma_{ci} \sigma_{c'i} W_i, \quad W_i = A d_i M_i^{-1} A d_i^T$$

with  $\sigma_{ci} \in \{-1, 0, +1\}$  the patch-body incidence and  $W_i$  the shared body's  $6 \times 6$  inverse **spatial** inertia transported to a common reference point, and  $G_{\text{patch}} = L$  exactly [h13 §1].

## 2.2 The conditions, each one measured

**(C1) One contact per body pair, or patch lumping.** Two contacts at the same interface with the same incidence sign produce an off-diagonal  $+(1/m_a + 1/m_b)$ . Measured on point contacts: max off-diagonal  $+2.0000e+0$  on the 3-box column, the 8-box K8 tower, the 3-box 1000:1 column and the  $2 \times 2 \times 2$  lattice — none of them a Laplacian [h13 §1A]. The 8-box tower fails **even with all rotation removed** [08 §1, table row 8]. After lumping,  $\|G_{\text{common}} - L_{\text{graph}}\|_{\infty} / \|G\|_{\infty} = 2.368e-17$  (3-box column),  $9.586e-17$  (K8 tower),  $2.368e-17$  (1000:1 column) [h13 §1B].

**(C2) Conductance = shared body inverse (spatial) inertia.** Scalar case:  $1/m_i$ . Block case:  $W_i = A d_i M_i^{-1} A d_i^T$ , symmetric positive definite,  $\min \text{eig } W_i = 7.99e-1 / 3.07e-1 / 7.99e-4 / 7.87e-1$  on the four scenes and  $\|W_i - W_i^T\| = 0.000e+00$  throughout; every off-diagonal block equals  $-W_{\text{shared}}$  to  $0.000e+00$  [h13 §1C].

**(C3) Normal angle.** In the scalar form the off-diagonal is  $-(1/m_i) \cdot n_c \cdot n_{c'}$ , so it turns positive past  $90^\circ$ : measured to survive  $89^\circ$  (max off-diagonal 0.0) and break at  $120^\circ$  (max off-diagonal  $5.0e-1$ ) [08 §1]. **The angle condition disappears under lumping:** on the lattice, patch normals  $90^\circ$  apart still give negative semidefinite blocks whenever the incidence sign product is  $-1$  (max  $\lambda^+ = -2.7986e-3$ ); the angle lives in the gauge  $\mathcal{D}$ , not in the conductance [h13 §1D].

**(C4) Rotation.** In the scalar form  $(r \times n)^T I^{-1} (r' \times n')$  has arbitrary sign and breaks the Laplacian; on a 3-box tower  $\|G_{\{n, t1\}}\|_{\infty} / \|G_{\{nn\}}\|_{\infty} = 0.1875$  [08 §1]. **Rotation does not break the block form:** it enters only through  $W_i$ , never through the incidence [h13 §1C].

**(C5) Friction rows do not break it.** For point masses with parallel normals,  $\|G_{nn} - G_{t1t1}\|_{\infty} = \|G_{nn} - G_{t2t2}\|_{\infty} = 0$  and all cross blocks are exactly 0, so  $G = L \otimes I_3$ : the tangential rows are two more independent copies of the same Laplacian [08 §1]. Friction changes the **cone**, not the operator — and the cone is what breaks the preconditioned iteration (§3.4, §6.1).

**(C6) Where it breaks: body degree  $\geq 3$ .** A body of degree  $d$  puts its  $d$  patches into a clique of the line graph, all with the same block conductance. The Laplacian sign condition needs  $\sigma_{ci} \sigma_{c'i} = -1$  for every pair at that body, and three or more  $\pm 1$  values cannot be pairwise opposite. Measured on the  $2 \times 2 \times 2$  box lattice: body degrees 4, 3, 4, 3, 4, 3, 4, 3; 16 edges with sign product  $+1$ ; worst off-diagonal block max  $\lambda^+ = +6.357e-01$  [h13 §1E]. Degree  $\leq 2$  — columns, towers, chains, and cycles of any length (Corollary C.2.1) — always admits the alternating orientation. The lattice is therefore a **signed** block Laplacian: still  $B W B^T$ , still PSD, still sparse and graph-assembled, but not an M-matrix, so effective-resistance machinery needs its signed variant there.

| condition                 | scalar form (point contacts)  | block form (patch-lumped)                  | measured                                                               |
|---------------------------|-------------------------------|--------------------------------------------|------------------------------------------------------------------------|
| one contact per body pair | required                      | not required (lumped)                      | +2.0e+0 off-diagonal without lumping [h13 §1A]                         |
| conductance               | $1/m_i$                       | $Ad_i M^{-1}_i Ad_i^T$                     | off-block = $-W_{\text{shared}}$ to 0.0 [h13 §1C]                      |
| normal angle              | $n_c \cdot n_{c'} \geq 0$     | irrelevant (absorbed in gauge)             | breaks at 120°, holds at 89° [08 §1]; 90° fine after lumping [h13 §1D] |
| rotation                  | must be absent                | absorbed in $W_i$                          | 0.1875 cross-coupling scalar [08 §1] vs 0.0 block [h13 §1C]            |
| friction                  | harmless, $G = L \otimes I_3$ | harmless (cone changes, operator does not) | cross blocks exactly 0 [08 §1]                                         |
| body degree               | $\leq 2$                      | $\leq 2$                                   | +6.357e−1 positive block at degree 3-4 [h13 §1E]                       |

### 2.3 Proof sketch

Write  $J = \sum_c e_c \sigma_{ci} Ad_i$  for the contact Jacobian in the common gauge, so that row block  $c$  of  $J$  picks the two bodies incident to contact  $c$  with opposite signs. Then

$$G = J M^{-1} J^T, \quad G[c, c'] = \sum_i \sigma_{ci} \sigma_{c'i} (Ad_i M^{-1}_i Ad_i^T) = \sum_i \sigma_{ci} \sigma_{c'i} W_i$$

which is the assembly rule of a weighted Laplacian on the graph whose vertices are contacts and whose edges are bodies — provided (a) each body contributes to at most two contacts, so that every off-diagonal is a single  $-W_i$  term rather than a sum of terms of mixed sign, and (b) the two incidences at a shared body have opposite signs, so that  $\sigma_{ci} \sigma_{c'i} = -1$  and the off-diagonal block is  $-W_i \leq 0$ . The diagonal then carries  $\sum_i \sigma_{ci}^2 W_i$ , and a body touched by an odd number of patches leaves a Dirichlet residue  $\sum_i \sigma_{ci} (\sum_{c'} \sigma_{c'i}) W_i$  on the row sum — the grounded form. Both statements are verified numerically, not assumed: column 3 of [h13 §1B] is the assembly agreement ( $\leq 9.6e-17$  relative), column 4 is the semidefiniteness of every off-diagonal block, column 5 is the Dirichlet row sum ( $\leq 4.8e-17$  on the three scenes that pass,  $1.9e-16$  including the  $2 \times 2 \times 2$  lattice, which does not).

The scalar form is the specialisation  $Ad_i = I_3$ ,  $M^{-1}_i = (1/m_i) I_3$ , at which point  $W_i$  commutes out and the off-diagonal becomes  $-(1/m_i) n_c^T n_{c'}$ , giving (C3). Condition (C6) is combinatorial: at a body of degree  $d \geq 3$  the pairwise products  $\sigma_{ci} \sigma_{c'i}$  over  $d$  values in  $\{-1, +1\}$  cannot all be  $-1$ .

The per-patch reference point and contact frame form a block-diagonal invertible gauge  $D_c = S_c \cdot \text{PERM} \cdot \text{blkdiag}(E_c, E_c) \cdot Ad(p_c - p_{\text{ref}})$ , and  $G_{\text{local}} = D G_{\text{common}} D^T$  holds to  $3e-15$  [h13 §1F], so  $G_{\text{local}}$  and  $G_{\text{common}}$  are congruent; the verdict of Theorem C.2 is read in the common gauge and does not transfer to the local one (Corollary C.1.1 and the counterexample there). The solve is done in the local gauge, where the cone is the normalised one. The gauge is also a free diagonal preconditioner:  $\text{cond}(G_{\text{patch}})$  over the non-zero spectrum is  $2.670e+03$  (unscaled local) vs  $3.494e+02$  (H13 gauge) vs  $3.059e+03$  (common) on the 3-box column, and  $2.627e+04 / 1.214e+04 / 8.817e+04$  on the K8 tower [h13 §1F].

The algebraic proof with stated hypotheses is Appendix C; §2.3 is the sketch it discharges.

### 2.4 Corollary 1 — the spectrum is the graph's

$\text{spec}(G) \setminus \{0\} = \text{spec of the generalized pencil } L_v x = \lambda M x$ , with  $L_v = J_{\text{common}}^T J_{\text{common}}$  the block **vertex** Laplacian of the contact graph,  $L_v[i, j] = \sum_c \sigma_{ci} \sigma_{cj} Ad_i^T Ad_j$ .

| scene                                 | $\max  \text{spec}(G) - \text{spec}(L_v, M) $ | relative           | source    |
|---------------------------------------|-----------------------------------------------|--------------------|-----------|
| masses 1/1/1 (point)                  | 0.0                                           | —                  | [08 §1]   |
| masses 1000/1/1000 (point)            | $6.4\text{e-}17$                              | —                  | [08 §1]   |
| masses 1/100/1000 (point)             | $3.7\text{e-}17$                              | —                  | [08 §1]   |
| 3-box column (patch)                  | $3.411\text{e-}13$                            | $6.510\text{e-}16$ | [h13 §1G] |
| 8-box K8 tower (patch)                | $1.251\text{e-}12$                            | $9.565\text{e-}16$ | [h13 §1G] |
| 3-box 1000:1 column (patch)           | $2.274\text{e-}13$                            | $4.836\text{e-}16$ | [h13 §1G] |
| $2 \times 2 \times 2$ lattice (patch) | $1.251\text{e-}12$                            | $1.115\text{e-}15$ | [h13 §1G] |

The lattice row matters: the spectral identity is  $K K^\top$  vs  $K^\top K$  with  $K = J \text{Minv}^{\{1/2\}}$ , which holds for *any*  $J$ , so the degree  $\geq 3$  condition of (C6) governs whether the **assembly** is a Laplacian and not whether the **spectrum** is readable [h15 §1A, §1D]. The mass-ratio bottleneck is therefore a generalized eigenvalue of the contact graph, computable without forming  $G$ .

A second identity in the same family: the grounded inverse satisfies  $G^{-1}_{\{ij\}} = (e_i - e_g)^\top L^+_{\text{aug}} (e_j - e_g)$  to  $3.2\text{e-}10$ , with  $\sum_e w_e R_e = 5.0000000000$  over the 5 edges of the augmented graph's 6 nodes (5 contacts plus the ground node  $g$ ), i.e. Kirchhoff's  $n - 1$  at  $n = 6$ , and per-contact leverage  $[1 \ 1 \ 1 \ 1 \ 1]$  — every contact in a chain is a bridge [08 §1].

## 2.5 Corollary 2 — patch lumping makes each patch block of $J$ full row rank, and on a loop-free scene

makes  $J_{\text{patch}}$  full row rank, so contact derivatives exist without regularisation

The lumping map  $C$  (corner impulses  $\rightarrow$  patch wrench) is the point at which static indeterminacy is quotiented out. On the pushed cube (0.20 m, 15 kg,  $\mu = 0.5$ , horizontal force 147.15 N, 4 coplanar corner contacts):

| quantity                                                                                   | point contacts                                    | patch                            | source    |
|--------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------|-----------|
| rank $J$                                                                                   | <b>6 of 12 rows</b>                               | <b>6 of 6 rows</b>               | [h13 §3A] |
| cond $G$                                                                                   | $\infty$                                          | $8.550\text{e+}00$               | [h13 §3A] |
| cond of the sensitivity system $\partial F/\partial \lambda$ resp. $\partial F/\partial w$ | $4.913\text{e+}16$ , singular                     | <b>7.259e+01</b> , non-singular  | [h13 §3C] |
| analytic derivative vs central FD                                                          | $7.261\text{e-}02$ (on $\lambda$ , min-norm pinv) | <b>1.886e-10</b> (on $w$ )       | [h13 §3C] |
| spread of the solution over 50 random warm starts                                          | $4.824\text{e-}01$ on $\lambda$ ( $\mu = 0.5$ )   | $2.211\text{e-}12$ on $C\lambda$ | [h13 §3B] |
| exact width of the point solution set, $\lambda_n(\text{corner } \theta)$ , $\mu = 0$      | $0.294300 \text{ N}\cdot\text{s}$                 | —                                | [h13 §3B] |

The 6-dimensional kernel of  $C$  **is** the self-stress of the coplanar patch.  $J_{\text{patch}}$  has full row rank, so  $G_{\text{patch}}$  is positive definite and the patch problem is strongly monotone with a unique solution. No null-space selection rule is needed because there is no null space [h13 §3A, §3C]. The normal+moment lumping is exact rather than a relaxation: the 3-dim patch cone ( $F_n \geq 0$ , centre of pressure inside the rectangle) is precisely the image of the 4-corner frictionless cone, measured onto and into with residual 0.0, and  $J_{\text{points}}^\top \lambda = J_{\text{patch}}^\top (C \lambda)$  to  $1.3\text{e-}16 \dots 2.0\text{e-}16$  relative [h13 §0].

What lumping conserves, measured on eight scenes: virtual work exactly ( $\|J_p^\top - J_w^\top C\|_\infty \leq 1.3878\text{e-}17$ ), rank exactly (18/18, 48/48, 66/66 rigid-body degrees of freedom), and the translation wrench to  $1.83\text{e-}10 \dots 5.22\text{e-}09$  [U89 §2; reproduced at  $1.38777878\text{e-}17$  and  $3.169\text{e-}10$  in A70 §4]. It restores the sign structure on every stack: the largest positive off-diagonal eigenvalue drops  $8.00 \rightarrow 0.0000\text{e+}00$  and the positive off-diagonal Frobenius share  $59\text{--}61 \ \% \rightarrow 0.00 \ \%$  [U89 §2]. It does not conserve the wrench under spin: 0.3232 relative on the rotating box and 0.2767

on pure spin [U89 §2; A70 §4]. Lumping is therefore a rank-restoring approximation, not a solution-preserving change of variables. The tangential part (Contensou ellipsoid + spin) **is** a model change; the measured gap is  $2.506e-05$  (K8 gravity) and  $3.359e-05$  (K8 rolling) in  $\|J^T \lambda_{\text{point}} - J_{\text{patch}}^T w\|_\infty / \|J^T \lambda\|_\infty$  [h13 §2B] and 0.3232 on the spin counterexample [U89 §2], which is not evidence that the two friction models agree in general.

The spin error is the patch cone’s capacity, not a lost moment:  $C$  has rank 6 and transports  $\tau_n$ , since tangential corner pairs give  $\tau_n = \sum (x_i f_{\{y,i\}} - y_i f_{\{x,i\}})$  [A70 §4]. Four corner Coulomb cones carry  $|\tau_n| = \mu F_n \sqrt{(e_1^2 + e_2^2)}$ ; the patch cone  $\|(F_{t1}, F_{t2}, \tau_n/\ell)\| \leq \mu F_n$  with  $\ell = \sqrt{(e_1^2 + e_2^2)/3}$  carries  $\mu F_n \ell$ . The ratio is  $\sqrt{3} = 1.7320508$ . Measured at  $F_n = 1$ ,  $\mu = 0.5$  on the square patch: point cone 0.07071068, patch cone 0.04082483 [A70 §4].

The same holds on the GPU:  $dw/d\mu$  against FD  $6.2e-9$  at  $\eta = 0$ , cond 72.6, GPU against the h13 CPU path  $2.3e-12$  [HUNT, U2]; after the retune,  $dw/d\mu$  against FD  $8e-9$  with full sha256 bit-identity [HUNT, U2b].

**Caveat.** This is the single-patch case. On the  $2 \times 2 \times 2$  lattice  $J_{\text{patch}}$  is still rank 48 of 96: lumping removes intra-patch indeterminacy, not loop indeterminacy. A closed lattice is hyperstatic at the wrench level too, and that is physics [h13 §1H, §3C].

## 2.6 Corollary 3 — block-tridiagonal exact solve on chains

For a column or a tower the patch line graph is a **path**, so  $G_{\text{patch}}$  is block tridiagonal (measured block bandwidth = 1 in every column/tower scene) and  $(G + \rho I)x = r$  is a  $6 \times 6$  block Thomas elimination — the chain case of Gaussian belief propagation, which is the same recursion as the articulated-body inverse inertia ( $\text{tree\_gls} \triangleq \text{ABA}$ : subtree Schur complement vs  $I^A$  agreeing to  $3.4e-16 / 1.9e-17 / 0$ ) [08 §3].

| scene                  | outer iterations, dense Cholesky | outer iterations, block-tridiagonal | max $ z_{\text{chol}} - z_{\text{tri}} $ |
|------------------------|----------------------------------|-------------------------------------|------------------------------------------|
| K8 tower, gravity      | 140                              | <b>140</b>                          | $1.67e-16$                               |
| K8 tower, rolling      | 1 848                            | <b>1 848</b>                        | $2.37e-15$                               |
| 1000:1 column, gravity | 296                              | <b>296</b>                          | $4.39e-12$                               |
| 1000:1 column, rolling | > 30 000                         | > 30 000                            | $1.22e+01$ (two non-solutions)           |

[h13 §2D]. The dense factorization is replaced by the graph solve with no change to the algorithm. No sketch is needed on a chain; a sketch is for the branched or lattice case where the elimination is not a path — and there the  $k = 64$  Spielman-Srivastava sketch is measured to be the wrong tool at these sizes (3–131 CG steps per ADMM iteration, CG cap hit on the asymmetric tower) [h11 §1B]. No nearly-linear-time Laplacian solver is run against the block Thomas elimination; that comparison is listed in §6.9. The sketch is run at  $k = 64$  only, which is below the  $O(n \log n / \epsilon^2)$  sample size the guarantee of [25] requires, so the conclusion “wrong tool at these sizes” is bounded to  $k = 64$ .

## 3. Consequences for solvers

### 3.1 Spectral $\rho$ from the pencil, in $\sim 200$ matvecs

The estimator runs matrix-free on  $K^T K$  with  $K = J \cdot \text{Minv}^{\wedge\{1/2\}}$ , Lanczos for  $\lambda_{\text{max}}$  and LOBPCG (block 8, 20 iterations, Jacobi preconditioner) for  $\lambda_{\text{min}}^+$ :

| scene | n_c | matvecs | $\lambda_{\text{min}}^+$ rel err | $\lambda_{\text{max}}$ rel err |
|-------|-----|---------|----------------------------------|--------------------------------|
| cube  | 4   | 10      | $9.9e-16$                        | $4.7e-16$                      |
| K8    | 32  | 204     | $1.2e-11$                        | $4.6e-16$                      |

| scene                        | n_c | matvecs     | $\lambda_{\min}^+$ rel err | $\lambda_{\max}$ rel err |
|------------------------------|-----|-------------|----------------------------|--------------------------|
| col rest / sliding / binding | 3   | 12 / 12 / 8 | 1.8e-12 ... 1.3e-13        | 2.2e-16 ... 0.0          |
| tower asym 1/1000            | 32  | 204         | 1.8e-9                     | 3.5e-15                  |
| 2×2×2 lattice                | 64  | 204         | 1.2e-15                    | 2.1e-7                   |

[h15 §1A]. The default budget is block  $8 \times 20 \approx \mathbf{6.4 \text{ matvecs per contact}}$ , and the Jacobi preconditioner is not optional: without it the asymmetric tower never converges (relative error 3.97 ... 57.2 in  $\lambda_{\min}$  at every budget from 46 to 364 matvecs, nine rows, no trend) [h15 §1C].

On the GPU the same estimate costs  $k_{\text{lanczos}} + \text{block} \cdot (\text{maxiter} + 1) \leq 204$  matvecs, **fixed in n\_c**, one pencil matvec being the same two sparse products as one CG matvec of the x-update. Measured cost share of one estimate against the solve it steers, in matvecs: **2.5 % (n\_c = 4 096), 6.0 % (1 024), 17 % (256), 35 % (asymmetric tower), 102 %** (the N = 10 000 lattice, where the solve is 25 iterations) [h1 §H1b.1]. The rule is: estimate once, cold; re-estimate only when the active set actually changed (an  $O(n_c)$  test); freeze  $\eta$  after the first estimate. Re-estimating at every Cadoux refresh is refuted — on the asymmetric tower 1 508 estimates = 307 632 matvecs = 664 Mflop, 13.91 s against 0.29 s, for **identical** iteration counts; the gated version keeps the whole benefit ( $29 \rightarrow 28$  and  $9 \rightarrow 8$  iterations) at 2 estimates [h15 §4].

**Cost against a dense pencil eigensolve.** Matrix-free Lanczos wins at packing scale and loses below it. On pack800 ( $n_c = 3\,022$ , pencil  $4\,800 \times 4\,800$ , 100 interleaved pairs with alternating method order): dense median 2 336.51 ms against Lanczos prefix medians 874.06 ms at tol  $1e-6$ , 1 072.23 ms at  $1e-8$  and 1 294.51 ms at  $1e-10$ , i.e. ratios of the medians 2.673×, 2.179× and 1.805×; U89 also reports median pairwise speedups of 2.90×, 2.40× and 2.00× for the same three pairs, which is a different summary measure. 100/100 runs inside tolerance [U89 §3]. Dense spans 1 083.08–4 216.82 ms and Lanczos at  $1e-6$  spans 277.4–1 624.2 ms on the same runs. The audit re-measured the same pair at tol  $1e-8$  and reports the median of the pairwise ratios: **2.692×** (dense 1 673.143 ms, Lanczos 611.307 ms, 100/100, pairwise ratios 1.252–3.536) [A70 §5]. On dem\_step ( $n_c = 441$ ) the dense solve wins: 59.07 ms against 669.96 ms at  $1e-8$  [U89 §3], 19.065 ms against 126.063 ms in the audit [A70 §5]. The claim is **2-3× at  $n_c \geq 3\,000$  and slower at  $n_c \leq 500$** . U89 §3 estimates a crossing between  $n_c \approx 1\,000$  and 1 500 from these two sizes; A70 §5 states that its own measurement fixes no crossing point. No size between 441 and 3 022 was measured. [MISSING: sizes between  $n_c = 441$  and 3 022] Every millisecond in this paragraph carries **GPU shared**. The earlier 12.6–84× reading is refuted: it ran Lanczos at 40 steps, which misses the  $1e-8$  gate at relative error  $9.63e-3$  and scores 0/100, and its 328 ms / 4 142 ms pair was not re-measured under matched load [A63 §“U67: tidsserien vid samma tolerans”; U89 §3].

### 3.2 $\rho = \lambda_{\min}^+$ , and what it is worth

ADMM iterations to natural-map residual  $1e-8$ ,  $\eta = 0$ , Cadoux outer loop, cold estimate (every contact assumed active, which is all a solver knows at iteration 0):

| scene          | n_c | $\rho = \lambda_{\min}^+$ | $\rho^* = \sqrt{(\lambda_{\min} \lambda_{\max})}$ | $\rho^*/10$ | $\rho = \text{mean}(\text{spec})$ | $\rho = \lambda_{\max}$ | $\rho = \text{median}\ G_{cc}\ _2$ | PGS       |
|----------------|-----|---------------------------|---------------------------------------------------|-------------|-----------------------------------|-------------------------|------------------------------------|-----------|
| cube           | 4   | 66                        | <b>60</b>                                         | 238         | 63                                | 86                      | 60                                 | <b>20</b> |
| K8             | 32  | <b>3</b>                  | 35                                                | 8           | 1 330                             | 4 100                   | 977                                | 14 682    |
| col rest       | 3   | <b>36</b>                 | 2 021                                             | 213         | > 5 004                           | > 5 004                 | > 5 004                            | 18 449    |
| col sliding    | 3   | <b>213</b>                | 2 520                                             | 216         | > 11 842                          | > 16 381                | > 12 832                           | 18 225    |
| col, set binds | 3   | <b>29</b>                 | 438                                               | 48          | 11 845                            | > 23 878                | > 15 833                           | 9 215     |
| tower asym     | 32  | <b>9</b>                  | > 5 052                                           | 1 512       | > 9 027                           | > 16 776                | > 7 809                            | > 30 000  |
| lattice        | 64  | <b>9</b>                  | 51                                                | 10          | 248                               | 568                     | 102                                | —         |

[h15 §2]. In this seven-scene comparison,  $\rho = \lambda_{\min}^+$  has the lowest count on 5 scenes, ties on one and loses 10 % on the cube. This subset is not a prediction result: §6.6 measures 2/21 hits for that rule and 5/21 for the declared structural selection. The rule  $\rho = \text{median}\|G_{\text{cc}}\|_2$  is reported here as a scalar step rule (977 against 3 on K8; non-convergence on all three columns and the asymmetric tower) [h15 §2], which is not how the Delassus diagonal is used in [16] and [17], where it scales the cone constraint. The block-diagonal Delassus scaling and block-Ruiz equilibration are run in §6.2 but not on these seven scenes. [MISSING: block-Delassus and block-Ruiz rows in this table] The result is checked three ways before it is believed: external anchors on the solution each rule produces ( $a_x = 4.905$  to  $1.95e-13$ ;  $\Sigma \lambda_n$  over 4 ground contacts =  $0.31392$  to  $1.95e-11$ ), tolerance scaling linear in the number of decades with no false floor, and agreement between rules on  $J^T \lambda$  to  $\leq 5.5e-9$  [h15 §2C].

The proximal term follows the same spectral reading:  $\eta = \alpha \cdot \lambda_{\min}^+$  caps the relative wrench error at  $\alpha$  on every scene ( $\text{err}/\alpha = 0.02$  on K8 and the asymmetric tower,  $0.29 \dots 0.42$  on cube and lattice,  $0.88$  on the 1000:1 column, constant in  $\alpha$ ), where the diagonal rule  $\eta_c = \alpha \|G_{\text{cc}}\|_2$  gives  $1.37 \dots 8.018$  and its ratio *grows* as  $\alpha \rightarrow 0$  [h15 §3; reproduced on the GPU in h1 §H1b.3].

### 3.3 Proximal ADMM with the exact cone: the headline numbers

| scene                                     | baseline                                             | with the structure                                                    | iteration-count ratio (not time) | source            |
|-------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------|-------------------|
| 1000:1 column, cone binds                 | PGS <b>18 225</b> sweeps                             | <b>216</b> ADMM iterations                                            | 84×                              | [h11 §1A]         |
| 1000:1 column, all active                 | PGS 18 449                                           | 213                                                                   | 87×                              | [h11 §1A]         |
| column of 8, 1000/1, cone binds           | PGS > 30 000 (res $5.2e-7$ )                         | 2 826                                                                 | —                                | [h11 §1A]         |
| K8 tower, patch-lumped                    | PGS <b>14 630</b> sweeps                             | patch block-GS <b>366</b> (40×), patch ADMM <b>140</b> (105×)         | 40× / 105×                       | [h13 §2]          |
| K8 tower, point ADMM                      | PGS 14 682                                           | <b>8</b> at $\rho^*/10$ , <b>3</b> at $\rho = \lambda_{\min}^+$       | 1 835× / 4 894×                  | [h11 §1A; h15 §2] |
| K8 tower, frictionless, exact block $L^+$ | 349 point sweeps                                     | <b>1</b> iteration                                                    | 349×                             | [h13 §2]          |
| asymmetric 1/1000 tower                   | PGS > 30 000; five methods did not converge [08 §1F] | <b>1 512</b> ( $\rho^*/10$ ) → <b>9</b> ( $\rho = \lambda_{\min}^+$ ) | 168× from 1 512                  | [h11 §1A; h15 §2] |
| 1000:1 column, frictionless               | 28 456 point-PGS sweeps                              | <b>1</b> (exact block $L^+$ )                                         | 28 456×                          | [h13 §2]          |
| 1000:1 column, friction, patch            | PGS > 30 000                                         | <b>296</b> ADMM                                                       | —                                | [h13 §2]          |
| pushed cube                               | PGS <b>20</b>                                        | ADMM 60                                                               | <b>0.33×</b> (loss)              | [h11 §1A]         |

The column divides PGS sweeps by ADMM iterations and is not a wall-clock ratio. Where both were timed single-threaded, one ADMM iteration costs  $0.09\times$  a PGS sweep at  $n_c = 32$  and  $0.82\times$  at  $n_c = 256$  [h11 §3]. The rows K8 tower, frictionless and 1000:1 column, frictionless divide a sweep count by one direct linear solve and are not an iteration ratio at all.

Two baseline numbers for the same K8 tower appear in the corpus — 14 630 [h13 §2, reproducing 08 §1F with  $\|G_{\text{points}} - \text{box\_tower}.G\|_\infty = 0$ ] and 14 682 [h11 §1A] — from the two lanes’ own scene constructors. Both are quoted with their source rather than merged.

Three properties of the ADMM row are worth separating, because they are usually conflated:

- **The Laplacian preconditioner is scene-dependent.** Counts are identical to dense Cholesky in the outer ADMM iteration; it changes the x-update cost. On the h11-b chain, Laplacian/Jacobi/no preconditioner take 1/3/3 median PCG iterations and 2.859/3.835/3.078

ms for 5×20 solves. On pack50 they take 62/35/33 iterations and 2.718/0.209/0.169 s: the Laplacian choice costs 1.77× the Jacobi iterations and 13.0× its time because  $\max |L-G_{nn}| = 286.4727554$  [A79 §2].

- **An ADMM iteration is cheaper than a PGS sweep:** 0.09× at  $n_c = 32$ , 0.82× at  $n_c = 256$ , single threaded — PGS needs  $n_c$  scalar root finds per sweep, ADMM one closed-form cone projection [h11 §3].
- **$\rho$  matters more than the preconditioner.** The spread over one decade either side of  $\rho^*$  is 4× to 90×, and  $10\rho^*$  fails outright on five of eight scenes [h11 §1C].

At scale and without contention, the same rule gives ADMM 84.09 ms (25 iterations) against PGS 445.61 ms (1 000 sweeps) on the  $N = 10\,000$  lattice with 40 000 contacts — **5.30×** [U20 §1]. The contended local figure for the same pair was 4.28× [U20 §1], and one previously published ratio does not survive the uncontended rerun: h1’s “ADMM 3.47× PGS at  $n_c = 256$ ” is a sweep-granularity artefact; the clean ratio is **0.38×** (PGS wins at that size) [U20 §1, §3.1].

### 3.4 The router, with the certificate first

The regime is readable **before the first ADMM iteration** by solving the bilateral problem  $\lambda_{bil} = -(G + \text{diag } \eta)^{-1} b$  on the active closed set (solve, drop every contact with  $\lambda_n \leq 0$ , repeat; 1–9 rounds on these scenes) and testing  $\lambda_{bil,c} \in \text{int } K_\mu$  per contact [h23 §9]. As a detector on the same 13 scenes it reaches AUC 0.944 (TPR 1.00 at FPR 0.11) against the ADMM-read signals, against AUC 0.917 for the cold one-solve version and AUC 0.889 for the normal-only signal; the operating threshold is selected on those 13 scenes and no held-out figure is reported [h23 §9]. [MISSING: held-out AUC]

Rules built on it, 13 scenes [h23 §9, TABLE 11]. The ladder is this paper’s own raise factor 1.15; the residual-balancing rule of [22, §3.4.1] and [23] at their published parameters ( $\mu = 10$ ,  $\tau^{\text{incr}} = \tau^{\text{decr}} = 2$ ) is not run. [MISSING: published residual-balancing baseline]

| rule                                                                   | converged / 13 | worst ratio to the per-scene optimum | median ratio |
|------------------------------------------------------------------------|----------------|--------------------------------------|--------------|
| bil_A (certificate $\rightarrow \rho_A$ , else $\rho_B + 1.15$ ladder) | 13             | 20.9                                 | <b>1.13</b>  |
| lad_A ( $\rho_A + \text{ladder}$ , no certificate)                     | 13             | <b>12.0</b>                          | 1.42         |
| ladder from $\rho_B$ (no structure read)                               | 13             | 16 368.0                             | 2.37         |

The full router solve(scene) — certificate  $\rightarrow$  exact bilateral solve | PGS + arc freeze | spectral ADMM | block-Ruiz ADMM — was then measured on 21 scenes spanning H1–H30 [U18 §1, §2]:

| statistic                                                     | value                                                           | source     |
|---------------------------------------------------------------|-----------------------------------------------------------------|------------|
| median ratio to the best of the four routed methods per scene | <b>1.00×</b>                                                    | [U18 §3.1] |
| worst ratio                                                   | 9.15× (a1_slip)                                                 | [U18 §3.1] |
| misrouted (> 3× that best)                                    | 4 of 21 (19.0 %)                                                | [U18 §3.1] |
| leave-one-out error                                           | 4 of 21 (19.0 %), the same four                                 | [U18 §3.1] |
| scenes solved by step 0 alone                                 | 7, at 1.0–6.3 sweep-equivalents, zero ADMM iterations           | [U18 §2]   |
| chains                                                        | 43 sweeps vs 18 225 plain PGS (423×)                            | [U18 §2]   |
| packings                                                      | 463–1 233 sweep-equivalents where PGS needs > 4 000 to > 20 000 | [U18 §2]   |

The router comparison now uses U135’s executed 25-point sweep oracle rather than U18’s hard-coded ORACLE\_MAP. U18’s table remains reproducible, but col8\_cone makes the router 0.106× the alleged oracle, so that table is not a per-scene optimum [A78 §2]. Against the executed baselines, PGS-100 reaches the  $1e-8$  tolerance on only 8/21 scenes; over all capped runs the router totals

52.4 s and PGS-100 totals 35.8 s, so the latter total includes 13 failures. On the two scenes where the router assigns ADMM  $\rho$ , its rule is  $1.6\times$  from U135's swept  $\rho^*$  on cube and  $3.1\times$  on a1\_slip; the other three audited scenes take a certificate or PGS branch [A78 §2, 4; U135 \$Tabell].

**The A1 regime.** Three of the four misrouted scenes are Unitree A1 quadruped instances: small contact count ( $n_c \leq 4$ ), high joint articulation, unanchored floating base. Plain PGS converges there in 27–53 sweeps while the multi-contact ADMM ladder spends 209–485 [U18 §3.3]. Adding the branch if is\_articulated and  $n_c \leq 4$ : PGS drops a1\_slip  $9.15\times \rightarrow 1.00\times$ , h25\_a1\_stick  $7.76\times \rightarrow 1.00\times$ , h25\_a1\_slip  $6.04\times \rightarrow 1.00\times$ , i.e. misrouted  $4/21 \rightarrow \mathbf{1/21 (4.8\%)}$ ; the generalized  $n_c \leq 4$  rule also takes the cube ( $3.01\times \rightarrow 1.00\times$ ) and gives **0/21** [U33 §1]. The caveat travels with the number: the rule was selected on the same 21 scenes and the leave-one-out figure for it is not reported [HUNT, U33]. [MISSING: LOO for the generalized  $n_c \leq 4$  rule]

## 4. What the spectrum is NOT

The spectral reading is a **conditioning** diagnosis. It was also proposed as a **stability** margin — if  $\lambda_{\min}^+$  of the active contact Laplacian falls toward zero, the stack is about to topple — and that proposal is refuted cleanly [U22].

**K8 tower under tilt and under lateral force.** Tilting  $0 \dots 7.125^\circ$  (the analytic toppling angle  $\arctan(0.10/0.80)$ ) and pushing  $0 \dots 39.24$  N (up to  $\mu\text{N}$ ),  $\lambda_{\min}^+$  is **identically constant** at  $2.3331\text{e-}03$  (point) and  $5.8329\text{e-}04$  (patch) all the way to toppling and to slip; Pearson  $r = 0.0000$ , Spearman  $\rho = 0.0000$ , and the warning given is **0 steps** [U22 §2.1, §2.2, summary lines 2–4].

**The certificate is the margin in the stated K8 model.** The bilateral certificate margin on the patch cone,  $\min_c (F_n - |M_{t2}|)/F_n$ , has the same zero crossing as the normalised ZMP distance and the Coulomb margin on every swept point [U22 §2.1, §5]. The reported  $r = +1.0000$  is not support: a monotone one-dimensional sweep gives  $r = 1$  by construction. On 40 randomized towers the correlation remains approximately 1 while the maximum pointwise certificate/ZMP discrepancy is 0.969, reported here as 0.97 [A81 \$Tal  $\times$  status  $\times$  verdikt]. The zero crossings, not the correlation, are retained. The mechanism is elementary once stated:  $G = J M^{-1} J^\top$  is a function of geometry and inertia only and is independent of the external load as long as the contacts stay in the same regime, while  $w_{\text{bil}} = -G^{-1} b$  reacts to the load through  $b$  [U22 §5].

**1000:1 column, mass ratio 1 ... 1e4.**  $\lambda_{\min}^+$  falls as a power law  $R^{-0.9792}$  from  $7.5758\text{e-}02$  to  $1.0141\text{e-}05$ , correlating with the condition number at Pearson  $r = -1.0000$ ; the physical toppling threshold moves by only 38 % ( $18.435^\circ \rightarrow 11.311^\circ$ ) while  $\lambda_{\min}^+$  falls  $7\,470\times$ , and the certificate margin stays 1.0000 (the stack rests in equilibrium at every ratio) [U22 §3]. So  $\lambda_{\min}^+$  is numerical stiffness, not distance to collapse.

**DEM packings under foot load.**  $\lambda_{\min}^+$  does not fall toward zero over the sampled load interval; it **rises** as the grains compress (pack200:  $1.4228\text{e-}02 \rightarrow 8.0689\text{e-}01$ ), and the correlation with distance to  $F^*$  is  $r = -0.6407 / \rho = -0.3929$ , non-monotone. The Terzaghi ratio  $F/F^*$  is an affine re-expression of the same force, so its  $r = +1.0000$  is definitional. The sweep reaches only  $0.27 F^*$  on pack200 and  $0.03 F^*$  on pack800, not the bearing limit [U22 §4, §5; A81].

**Verdict, stated as the negative it is:** the spectrum carries conditioning; the certificate carries stability [U22 §5].

## 5. Load ordering from effective resistance

If  $G$  is a Laplacian, the load path is a current. Injecting a dipole at the loaded body and grounding at the support, the edge current  $|I_e|$  is a candidate predictor of the contact force  $|\lambda_e|$ .

Measured on 8 scenes,  $N = 50 \dots 63\,365$  grains,  $n_c = 176 \dots 124\,311$  [U21 §1]:

| predictor                             | Spearman $\rho$ , fine packings (dw8/dw16) | Precision@10 %, pack50 | source         |
|---------------------------------------|--------------------------------------------|------------------------|----------------|
| resistance flow (H13 conductance 1/m) | <b>+0.390 ... +0.401</b>                   | <b>0.444</b>           | [U21 §1, §3.1] |
| contact degree                        | +0.322 ... +0.339                          | 0.111                  | [U21 §1, §3.1] |
| distance to load                      | +0.110 ... +0.149                          | 0.278                  | [U21 §1, §3.1] |
| random                                | −0.001 ... +0.036                          | 0.056                  | [U21 §1]       |

The margin over contact degree is 0.06 in Spearman. Changing the conductance model moves the same statistic by 0.038 (0.3898 / 0.3651 / 0.3515 on dw16) [U21 §3.2]. No confidence interval is reported for any entry of the table, and the two spans overlap.

Three qualifications, all measured:

- **The conductance model barely matters.** H13's 1/m, unweighted, and Hertz-stiffness weighting agree to within 0.02 in Spearman (dw16: +0.3898 / +0.3651 / +0.3515). Connectivity and boundary conditions dominate, not the local scalar weights [U21 §3.2].
- **Local top-5 % detection under the foot is not won.** Plain distance to the foot centre reaches Precision@5 % = 0.365 (dw4) and 0.267 (dw8) against the resistance flow's 0.286 and 0.142: directly under the penetrator, proximity is the whole story [U21 §3.1].
- **The sketch fails on a directed dipole.** ResistanceSketch (Spielman-Srivastava,  $k = 64$ ) falls to Spearman −0.079 ... +0.145 from projection noise, while sparse MINRES solves the same exact flow in 0.44 s at  $N = 63\,365$  (124 311 edges) [U21 §3.3].

**The grain-scale law is the graph's.** U7 measured a DEM limit load  $F^* = C \cdot d^p$  with  $p = 1.22$  [1.07, 1.38] for quartz and 1.16 for the proxy, extrapolating to 0 as  $d \rightarrow 0$  against a Terzaghi prediction of 56–66 N [HUNT, U7]. The flow graph's spectral gap is fitted on the same four grain sizes:

| quantity                                                | fitted exponent $p$ in $Y = C \cdot d^p$ | source     |
|---------------------------------------------------------|------------------------------------------|------------|
| $F^*$ (quartz limit load, this fit)                     | 1.133                                    | [U21 §2]   |
| $F^*$ (U7 reference)                                    | 1.22 [1.07, 1.38]                        | [HUNT, U7] |
| $\lambda_{\min}(L_{\text{grounded}})$ of the flow graph | <b>1.262</b>                             | [U21 §2]   |
| $\lambda_{\min^+}(L_{\text{sub}})$                      | <b>1.277</b>                             | [U21 §2]   |

No confidence interval is reported for 1.133, 1.262 or 1.277. The rank correlation  $\lambda_{\min}(L_{\text{grounded}})$  vs  $F^*$  is Spearman +0.8000 (log-log Pearson +0.6660) over  $n = 4$  points,  $d = 40 / 20 / 10 / 5$  mm and  $N = 124 / 990 / 7\,921 / 63\,365$  grains [U21 §2], which does not separate from chance at any conventional level. The two exponents are reported side by side; no claim that they are the same exponent is made. Read together with §4: the spectrum does not measure how close **this** packing is to failure, but it does carry how bearing capacity scales with grain size.

## 6. Limitations

**6.1 The cone metric, not the preconditioner.** The exact inverse solves in one iteration whenever the solution is strictly interior — including at  $\mu = 0.5$ , so friction per se is not the problem — and fails at 30 000 the moment any constraint binds, because the projection then happens in the wrong metric. Measured before lumping [08 §1E: sketched  $L^+$  at  $k = 16/64/256$  and the exact inverse, all  $> 30\,000$  with the cone binding] and after [h13 §2C: exact block  $L^+$  1 iteration interior,  $> 30\,000$  on both rolling scenes]. The preconditioner's home is the ADMM x-update, where the matrix is a graph Laplacian; a cone-compatible preconditioner for a projected method was not built.

**6.2 The seven-scene comparison does not establish a  $\rho$  rule; packings require  $\eta$ .** The seven scenes of §3.2 include cases where  $\rho = \lambda_{\min}^+$  has the lowest count among the tested settings, including 9 iterations against 51 at  $\rho = \sqrt{(\lambda_{\min} \lambda_{\max})}$  on the  $2 \times 2 \times 2$  lattice [h15 §2]. The 21-scene sweep in §6.6 supersedes that subset as an evaluation of prediction and gives 5/21 within  $2 \times$  for the declared structural choice. On the frozen DEM step, ADMM stalls at  $4.8e-3$  with  $\rho = \lambda_{\min}^+$  and converges at  $\rho = \rho^* = 46.5$  (12 CG, 800 iterations,  $F_z = 93.82$  N, bit-identical) [HUNT, H1b]. Both candidates come out of the same estimate and  $\text{cond}(G)$  does not separate them [HUNT, H1b]. The degree rule of §2.2 does not rescue the choice either: H18 tested  $\rho$  from graph degree and it fails on 7 of 12 scenes, with  $\rho_B$  costing  $17.5 \times / 1.577 \times / 5.7 \times$  on K8, the asymmetric tower and the lattice; no single-covariate threshold works (best  $n_c \leq 106$ , 3/12 LOO errors); and on `dem_step/pack200/pack800` **neither** candidate converges [HUNT, H18]. Packings need compliance: at  $\eta = 0$  nothing converges and the residual parks at  $5e-7 \dots 4e-5$  [HUNT, H18]. What does work there is structural but not spectral — block-Ruiz equilibration plus a constant  $\rho = 0.1$  wins the packing class (pack800  $0.59 \times$ ) without any spectrum, because after block-Ruiz all  $\eta > 0$  scenes share a spectrum; per-row Ruiz is inadmissible (it breaks cone isotropy) and the rule does not transfer to chains [HUNT, H23b].

And  $\eta$  is not free. A single global  $\eta$  is wrong:  $\eta = 1e-2$  costs the cube 3.6 % of its wrench and the 1000:1 column **95 %** [h11, Verdict]. The per-contact repair  $\eta_c = \alpha \cdot G_{cc}$  does not work either where it is most needed: against the external anchor  $F_n(\text{ground}) = M g \Delta t = 39.3185$ ,  $\eta = 0$  hits it to  $2.3e-6$  while  $\eta = 1e-2$  sits 94 % low, and the cost law  $\eta/(\eta + G_{cc})$  predicts  $1e-4$  where 0.231 is measured ( $2.300 \times$ ), because  $\min \text{diag}(G) = 1.0$  while  $\lambda_{\min}(G) = 2.5e-5$  — the soft global mode of a 1000:1 stack is invisible to every contact diagonal [h13 §5D]. The tested compliance  $\eta = \alpha \cdot \lambda_{\min}^+$  controls that error in the reported case (§3.2); this is a compliance result, not evidence for spectral  $\rho$  prediction [h13 §5E].

**6.3 Lattices.** Body degree  $\geq 3$  turns the assembly into a signed block Laplacian (§2.2, C6). The consequences are measured and partly benign: the spectral identity survives (the lattice is estimated as exactly as the chains,  $1.2e-15$ ) [h15 §1A]; the absolute-value fallback that would force a true Laplacian is **31  $\times$  wrong** ( $\lambda_{\min}^+$  59 % low,  $\lambda_{\max}$  36 % low,  $\rho$  49 % wrong) [h15 §1E]; effective resistance needs its signed variant [h13 §1E]; and  $J_{\text{patch}}$  stays rank deficient (48 of 96), so loop indeterminacy — and with it the derivative question — remains open on closed lattices [h13 §1H]. Separately, the  $6n_p$  block-Laplacian side of the estimator fails on the lattice for a scaling reason, not a sign reason: its Jacobi-preconditioned condition number is  $80 \times$  to  $28\,000 \times$  worse than the vertex pencil's and LOBPCG stalls at every budget tried; use the vertex pencil [h15 §1D, §1E].

**6.4 The A1 fourth regime.** Small- $n_c$  floating-base articulated robots are a regime the three graph-structural classes do not cover: PGS sweeps cost microseconds and converge in 27–53 sweeps where the ADMM ladder over-solves the dual variables for 209–485 [U18 §3.3]. This is the class on which the router's worst ratio ( $9.15 \times$ ) sits before the branch is added [U18 §3.1], and the branch that fixes it is a kinematic test (`is_articulated` and  $n_c \leq 4$ ), not a graph-spectral one [U33 §1]. A related datum: on A1 the scene wants  $\rho \approx 0.15\text{--}0.20 \lambda_{\max}$ , and  $\lambda_{\min}^+$  diverges to an all-open branch [HUNT, H30]; and the  $\rho$  ladder there climbs  $\rho_B \rightarrow 0.255 \lambda_{\max}$  over 11 raises, 476 iterations, where fixed  $\rho_A/\rho_B$  hit the 6 000 cap [HUNT, U9].

**6.5 The GPU x-update: the PCG gap.** The CPU results above use a dense Cholesky x-update; the GPU uses preconditioned CG, and the accuracy of that inner solve is part of the  $\rho$  rule, not an implementation detail. On the asymmetric tower the CPU needs 9 iterations at  $\rho = \lambda_{\min}^+$ , and the GPU needs 9 at  $\text{cg} = 64$ , **210 at  $\text{cg} = 32$ , and diverges at  $\text{cg} = 16$**  — “ $\rho = \lambda_{\min}^+$  with an x-update solved to the accuracy  $\rho$  demands” [h1 §H1b.2]. Isolated across  $\text{CG} \in \{16, 64, 256, \text{exact device Cholesky}\}$  [U33 §3]:

| scene               | finding                                                                                                                                                    | source     |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| col_slide (n_c = 3) | outer iterations 76 regardless of CG count and under exact Cholesky; exact device Cholesky 42.88 ms = 3.3× faster than CG = 16, 44.2× faster than CG = 256 | [U33 §3.1] |
| pack50 (n_c = 176)  | exact device Cholesky reproduces the CPU H23 count identically (540 vs 540); bil_A converges in 436                                                        | [U33 §3.2] |
| pack200 / pack800   | CG = 16 converges in 436 / 628, within 1.09× of CPU Ruiz (565 / 577); CG = 64/256 add 1.4–2.7× wall time and no fewer outer iterations                     | [U33 §3.2] |

The operating rule that closes the gap: CG = 16 for packings with  $n_c > 64$ , exact device Cholesky for  $n_c \leq 64$  [U33 §3.2]. A related GPU-side red line is closed separately: the A1 probe’s  $d\lambda/db$  error was PCG residual error in B amplified by  $(1 - \rho(B))^{-2} = 150$  at  $\rho(B) = 0.918$ ; Neumann defect correction ( $7.1e-13$ ) or a dense device LU at  $n_c \leq 4$  ( $4.5e-12$ ) fixes it [HUNT, U42]. Ruiz on GPU is still red: no  $\eta > 0$  packing converges through h1’s 16–64 PCG x-step where h23’s exact Cholesky does [HUNT, U9].

**6.6 The  $\rho$  rule in the 21-scene sweep: 5 of 21.** The category rule — rule A,  $\rho_A = \lambda_{\min}^+$ , when  $C1 = 1$ , and rule B,  $\rho_B = \sqrt{(\lambda_{\min}^+ \lambda_{\max})}$ , otherwise — was previously scored against a historical oracle table. That table is withdrawn as a reference: on 10 of its 21 rows the oracle equals  $\rho_{\text{pred}}$  to within  $1.1e-4$  relative (K8, col\_rest, tower\_asym, lattice, rand256, col\_heavy\_top, al\_stick, al\_slip, al\_h30, humanoid), on pack800 it equals  $2.0000 \times \rho_{\text{pred}}$  — U83 §1.2 names the cause itself, “Avrundning i orakeltabell ( $503.8 = 2.0 \times \rho_B$ )” — and col8\_cone carries K8’s value  $2.3330e-03$  while h25\_al\_stick and h25\_al\_slip carry al\_stick’s value  $1.6835e-02$  [U83 §1.1, §1.2; A70 §6, §7]. Eleven of the 21 rows therefore score the prediction against itself. The “18 of 21 within  $2\times$ ” of the previous draft is withdrawn.

U135 measures a 25-point grid  $\lambda_{\min}^+ \times \text{logspace}(-3, 3, 25)$  on all 21 scenes. Each point starts from zero, uses the same ADMM/Cadoux iteration and stops at natural-map residual below  $1e-8$  or at 5 000 iterations. The operator is  $G = J M^{-1} J^T + \text{diag}(\eta)$ , with U33’s contact compliance on dem\_step and the three packing scenes and zero compliance elsewhere.  $\lambda_{\min}^+$  is the first eigenvalue above  $1e-11$   $\lambda_{\max}$ . Among converged points,  $\rho^*$  is the lowest-iteration grid point, with the lowest  $\rho$  breaking ties [U135 §§Operator och lösare, Klass och träffdefinition].

The class column is U89’s C1 and body degree recomputed from the same J; all 21 agree with U89’s corrected columns [U135 §Klass och träffdefinition]. A hit has  $\max(\rho^*/\rho_{\text{pred}}, \rho_{\text{pred}}/\rho^*) \leq 2$ . Rule A hits **2/21**, rule B **7/21**, and the declared structural selection A if  $C1 = 1$ , otherwise B, hits **5/21**. Selecting the better ratio of A and B after observing the sweep hits **8/21**; this is an after-the-fact selection, not a prediction. The spectral rule is therefore not used as the router’s primary choice.

Table 6 — spectral  $\rho$  predictions against the measured grid minimum on 21 scenes [U135 §Tabell; build/U135/tabell6\_v2.csv]. Ratio is  $\rho^*/\rho_{\text{pred}}$ ; within  $2\times$  uses the reciprocal-symmetric ratio.

| scene | class from U89 | $\lambda_{\min}^+$ | $\lambda_{\max}$ | $\rho^*$        | iterations | $\rho_A$       | $\rho_B$      | $\rho^*/\rho_A$ | $\rho^*/\rho_B$  | structural rule | $\leq 2\times$ |
|-------|----------------|--------------------|------------------|-----------------|------------|----------------|---------------|-----------------|------------------|-----------------|----------------|
| cube  | C1=0; degree=4 | 0.11169 6312       | 0.95497 03547    | 0.19862 72518   | 58         | 0.11169 6312   | 0.32659 86324 | 1.77827 941     | 0.60816 92698    | B               | yes            |
| K8    | C1=0; degree=8 | 0.00233 314724     | 46.3793 3351     | 2.33314 724e-06 | 2          | 0.00233 314724 | 0.32895 26014 | 0.001           | 7.09265 4777e-06 | B               | no             |

| scene         | class from U89   | $\lambda_{\min}^+$ | $\lambda_{\max}$ | $\rho^*$        | iterations | $\rho_A$        | $\rho_B$       | $\rho^*/\rho_A$ | $\rho^*/\rho_B$ | structural rule | $\leq 2 \times$ |
|---------------|------------------|--------------------|------------------|-----------------|------------|-----------------|----------------|-----------------|-----------------|-----------------|-----------------|
| col_rest      | C1=1; degree=2   | 0.000292768252     | 2.00100025       | 2.92768252e-07  | 4          | 0.000292768252  | 0.02420391178  | 0.001           | 1.209590642e-05 | A               | no              |
| col_slide     | C1=1; degree=2   | 0.000292768252     | 2.00100025       | 0.0009258145028 | 162        | 0.000292768252  | 0.02420391178  | 3.16227766      | 0.03825061466   | A               | no              |
| col_open      | C1=1; degree=2   | 0.000292768252     | 2.00100025       | 0.001646356868  | 27         | 0.000292768252  | 0.02420391178  | 5.623413252     | 0.06802028048   | A               | no              |
| tower_asy     | C1=0; degree=8   | 3.77871314e-06     | 24.02119446      | 3.77871314e-09  | 2          | 3.77871314e-06  | 0.009527287292 | 0.001           | 3.966200477e-07 | B               | no              |
| lattice       | C1=0; degree=16  | 0.4021119685       | 65.15994714      | 0.0004021119685 | 2          | 0.4021119685    | 5.118749321    | 0.001           | 7.855668314e-05 | B               | no              |
| rand256       | C1=0; degree=13  | 0.2163405649       | 25.76148555      | 3.847139722     | 202        | 0.2163405649    | 2.360774097    | 17.7827941      | 1.629609426     | B               | yes             |
| dem_step      | C1=0; degree=14  | 0.6172496689       | 526.1260007      | 61.72496689     | 1104       | 0.6172496689    | 18.0208518     | 100             | 3.425196964     | B               | no              |
| pack50        | C1=0; degree=10  | 4.935228954        | 4306.254112      | 493.5228954     | 577        | 4.935228954     | 145.7818575    | 100             | 3.385351949     | B               | no              |
| pack200       | C1=0; degree=10  | 4.870694009        | 4386.170694      | 487.0694009     | 833        | 4.870694009     | 146.1632489    | 100             | 3.332365726     | B               | no              |
| pack800       | C1=0; degree=11  | 4.773906827        | 4759.045125      | 477.3906827     | 926        | 4.773906827     | 150.7290218    | 100             | 3.167211443     | B               | no              |
| rand256_h21   | C1=0; degree=256 | 126.7338501        | 2889.305391      | 225.3681962     | 800        | 126.7338501     | 605.1221334    | 1.77827941      | 0.3724342306    | B               | no              |
| col_heavy_top | C1=1; degree=2   | 0.000292768252     | 2.00100025       | 0.01646356868   | 2681       | 0.000292768252  | 0.02420391178  | 56.23413252     | 0.6802028048    | A               | no              |
| col8_cone     | C1=1; degree=2   | 7.602904289e-05    | 2.001707419      | 0.01352008815   | 3027       | 7.602904289e-05 | 0.01233644597  | 177.827941      | 1.095946773     | A               | no              |
| a1_stick      | C1=0; degree=4   | 0.01683484815      | 5.586013882      | 1.683484815e-05 | 2          | 0.01683484815   | 0.3066589237   | 0.001           | 5.489763008e-05 | B               | no              |
| a1_slip       | C1=0; degree=4   | 0.01683484815      | 5.586013882      | 0.946693082     | 627        | 0.01683484815   | 0.3066589237   | 56.23413252     | 3.087120605     | B               | no              |
| humanoid      | C1=0; degree=2   | 0.05780543824      | 0.5933695941     | 5.780543824e-05 | 3          | 0.05780543824   | 0.1852025632   | 0.001           | 0.0003121200767 | B               | no              |
| a1_h30        | C1=0; degree=4   | 0.2721286926       | 7.610089619      | 2.721286926     | 56         | 0.2721286926    | 1.439070443    | 10              | 1.891003279     | B               | yes             |
| h25_a1_stick  | C1=0; degree=4   | 0.01898863344      | 5.586100305      | 0.1898863344    | 200        | 0.01898863344   | 0.3256875973   | 10              | 0.5830321326    | B               | yes             |

| scene       | class from U89 | $\lambda_{\min}^+$ | $\lambda_{\max}$ | $\rho^*$     | iterations | $\rho_A$      | $\rho_B$     | $\rho^*/\rho_A$ | $\rho^*/\rho_B$ | structural rule | $\leq 2\times$ |
|-------------|----------------|--------------------|------------------|--------------|------------|---------------|--------------|-----------------|-----------------|-----------------|----------------|
| h25_al_slip | C1=0; degree=4 | 0.01756848743      | 5.586491561      | 0.5555643533 | 248        | 0.01756848743 | 0.3132829501 | 31.6227766      | 1.773362876     | B               | yes            |

The sweep contains 525 grid points per run. Of these, 223 reach the 5 000-iteration cap and 17 end in a numerical failure; failures are not replaced by historical oracle values. Six scenes have a boundary minimum, so those rows identify only a grid result, and six scenes have tied minima; the table consistently chooses the lowest  $\rho$  [U135 §§Klass och träffdefinition, Censurering, numerikfel och delade minima]. Two independent process runs match byte for byte on all 525 solutions, and U33’s dense solver matches the winning iteration count on 21/21 scenes [U135 §§Kontroll mot U33, Bitidentitet]. No prefactor is fitted here.

**6.7 The slip-saturated case.** A failed case for the structural rule occurs where the active set saturates the friction cone. On col8\_cone — 8 contacts, 7 of them sliding, mass ratio 1000:1 —  $\rho = \lambda_{\min}^+ = 7.6029\text{e-}05$  against a reference value of  $2.3330\text{e-}03$ , a ratio of **30.6856 $\times$**  [U83 §1.1]. That reference value is not a sweep on col8\_cone: it is numerically K8’s  $\lambda_{\min}^+ 2.3331\text{e-}03$  on the same historical table [U83 §1.1, row K8]. U135 instead measures  $\rho^* = 1.352008815\text{e-}02$ , 177.827941 times  $\rho_A$  and 1.095946773 times  $\rho_B$ ; the structural rule selects A and misses the  $2\times$  criterion [U135 §Tabell]. The historical 30.6856 $\times$  is a ratio against another scene’s spectrum and is not the measured oracle ratio. Restricting the spectrum to the sliding contacts moves  $\lambda_{\min}^+$  to  $7.6066\text{e-}05$  and the ratio to 30.67 $\times$ , i.e. the lowest eigenvalue is set by the mass distribution, not by tangential coupling [U83 §2.2]; block-Ruiz before the estimate leaves 23.59 $\times$  [U83 §2.4]. The rule underpredicts  $\rho^*$  by 28–31 $\times$  in this regime [U83 §5]. Holding the linear spectrum fixed does not hold  $\rho^*$  fixed: at  $M = 10$  with  $\lambda_{\min}(G) = 6.7911\text{e-}03$ ,  $\rho^*$  runs  $5.1603\text{e-}01$  at  $\mu = 0.1$  to  $3.9806\text{e-}02$  at  $\mu = 1.0$ , a factor **13.0 $\times$**  [U87 §5.1]. These results do not define a scene-level spectral  $\rho$  predictor.

What replaces the spectrum there is a theorem for one contact. For the ADMM map with the de Saxcé shift  $\Gamma(Gz + b)$  refreshed every iteration, Euclidean cone projection, no over-relaxation and  $G = gI$ , a fixed point with positive normal impulse and positive slip speed has the linearised spectrum  $\{0, 0, 0, g/(g+\rho), (\rho + g\mu^2/(1+\mu^2))/(g+\rho), (\rho^2 + gk)/((g+\rho)(\rho+k))\}$  with  $k = v/(\mu l)$ , and the spectral radius has a single minimum:

$$\rho_{\text{opt}} = g / (1 + \mu^2) , \quad q_{\text{opt}} = (1 + \mu^2) / (2 + \mu^2) .$$

The derivation is in [U95 §3] and is re-derived independently in [A68 §1]; it is not reproduced in Appendix C, which covers the theorem of §2 and its corollaries only. [MISSING: the one-contact derivation in this paper] Checks: the spectral formula against the matrix eigenvalues  $1.332268\text{e-}15$  and  $\rho_{\text{opt}}$  against numerical minimisation  $6.119971\text{e-}09$  [U95 §8]; the audit’s own Jacobian against U95’s  $2.220446\text{e-}16$ , central FD of the map  $2.186605\text{e-}10$ ,  $\rho_{\text{opt}}$  relative error  $3.360987\text{e-}08$  [A68 §1]. With the shift frozen the optimum moves to  $\rho = g$ ,  $q = 1/2$  [U95 §3; A68 §1]. The statement is about the local asymptotic rate: it does not fix the number of steps from a cold start, where the grid winner sits at 0.764–1.018 of the formula over 30  $(g, \mu)$  points and the exact theory point costs more steps than the grid winner on 15 of those 30 at tol  $1\text{e-}10$  [A68 §2]. The empirical form  $1/(\mu(1+\mu^2))$  proposed earlier is withdrawn: the constant it would require varies by 4.25 $\times$  over ten  $\mu$  points, the ratio between the two forms is  $\mu$ , and 5 of the 12 candidate points behind it are budget stops rather than measured minima [U95 §4, §6, §7].

The limit for two contacts is a slip-direction dependence, not an  $n_c$  law. For  $G = L \otimes I$  with equal  $\mu$  and parallel slip directions the two branches cross at  $\rho_{\text{opt}} = (-\beta a + \sqrt{(\beta^2 a^2 + 4ab/(1+\mu^2))})/2$ ,  $\beta = \mu^2/(1+\mu^2)$ ,  $a = \lambda_{\min}$ ,  $b = \lambda_{\max}$ ;  $\rho_{\text{opt}}/\sqrt{(ab)}$  depends on  $\kappa = b/a$ , so no factor of  $\mu$  alone suffices even inside that class [U95 §6; A68 §3, max relative error  $1.580023\text{e-}08$  on six coupled parallel

cases]. With  $G$  and its spectrum unchanged, turning the two slip directions from parallel to opposite moves  $\rho_{\text{opt}}$  from **6.825486 to 53.027909** at  $\kappa = 100$ ,  $\mu = 1$  [A68 §3]. No generalisation to  $n_c$  contacts is stated.

Mixed stick/slip active sets carry a second obstruction. On col8 at  $M = 1000$ ,  $\mu = 0.5$  the fixed point has 2 sliding contacts and 6 at zero slip speed; coordinate-wise central FD gives  $4.2e-09$  while the directional residual is 62.26 at  $h = 1e-06$ , so the central-difference matrix is not a Fréchet Jacobian there [U95 §5]. On a minimal two-contact scene with one strict-stick and one sliding contact the one-sided directional derivatives differ by 0.511889 at every step from  $1e-02$  to  $1e-07$ , while the frozen-shift control goes to zero [A68 §5]. This is a counterexample to differentiability at that state, not an impossibility result for every  $\rho$  rule, and it is not quoted as one [A68 §5, §6]. The router sends these scenes to PGS or to an exact device Cholesky/LU x-update rather than to a spectral  $\rho$  [U33 §1, §3.1; U55 §§1, 5].

**6.8 Coverage of the evidence.** All seven sources previously listed as unaudited now have independent audits. A77 audits h13: the rank-6 wrench map and polygon normal cone stand for non-collinear vertices, while frictional spin equivalence falls. A78 audits U18: branch reproduction and bitidentity stand, the hard-coded oracle claim falls, the PGS-100 comparison stands only for 8/21 converged scenes, and the  $\rho$  rule is  $1.6\text{--}3.1\times$  from the swept optimum where used. A79 audits h11: the solver counts stand, 216 belongs to  $\rho^*/10$ , and the preconditioner result is scene-dependent. A80 audits h15: spectrum,  $\rho$ ,  $\eta$ , lattice and gauge claims stand, with method-independence and timing caveats. A81 audits U21 and U22: U21’s benchmark stands as scene-bound descriptive data;  $r = 1$  is non-diagnostic, the 40-tower pointwise deviation is 0.97, and the DEM sweep reaches 0.27/0.03  $F^*$ . A82 audits 08 §1: its cited numerical quantities stand, with the scalar identity additionally requiring dynamic-body degree at most two and opposite incidence signs [A77–A82]. Earlier audits retain their outcomes: A63 withdrew U67’s figure CSV, A68 fell U95’s iteration-count reading, A70 restricted U89, A73 bounded U131 to a constant step load, and A74 identified U134’s measured quantity as the slip limit.

**6.9 Comparisons that are named and not run.** Each item below is a baseline a referee will ask for. None is measured in this paper and no number in this subsection is new.

- The convex relaxation of Anitescu-Potra and the solvers built on it (APGD, projected Gauss-Seidel over the convex cone) is dismissed in one clause of §1 (“rather than its convex relaxation”) and is never run on these scenes [37]. [MISSING: convex-relaxation iteration counts on these scenes]
- The adaptive- $\rho$  rule of [17] — ADMM + proximal with hyperparameter adaptation — is the direct competitor of every  $\rho$  choice argued for in §3.2, §3.3 and §6.6, and it is not run on these scenes. [MISSING: [17]’s adaptive  $\rho$  on these scenes]
- No nearly-linear-time Laplacian solver (Spielman-Teng 2004; Koutis-Miller-Peng 2011; Kyng-Sachdeva 2016) is run against the block Thomas elimination of §2.6 or the CG x-update of §6.5; the constant factors of those methods at  $n_c \leq 3\,022$  are not measured here, and none of those three works carries a verified bibliographic entry in the reference list. [MISSING: nearly-linear Laplacian solver baseline]
- No run of the Acary-Cadoux-Lemaréchal-Malick formulation [5] or of Cadoux’ algorithm [6, 7] on these scenes is included; the outer loop named after Cadoux is this paper’s own implementation. [MISSING: baseline runs of [5], [6], [7]]

## 7. Reproducibility

**Bit identity.** Every headline row is produced by a deterministic solver and checked across independent cold runs with full sha256 digests of the solved impulses:

| lane                | claim                                                                                                                          | source         |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------|
| U18 router          | <b>21 of 21</b> scenes bit-identical, two independent cold runs, CPU float64, full digests in <code>r_bits.json</code>         | [U18 §4]       |
| U33 GPU router      | <b>21 of 21</b> bit-identical, <code>fuse_fp=False</code> , <code>fast_math=False</code> , full digests                        | [U33 §4]       |
| U21                 | benchmark sha256 d07af481..., spectral sha256 6fb467d5..., two runs identical                                                  | [U21 §4]       |
| U22                 | three experiments, sha256 7a67454a..., 7d95b4aa..., 8b0af8a4..., PASS                                                          | [U22 §1]       |
| h1b spectral $\rho$ | $\lambda$ bit-identical over two gated runs on every scene                                                                     | [h1 §H1b.2]    |
| h13, h15, h11       | float64 CPU, numpy/scipy only; $\ G_{\text{points}} - \text{box\_tower}.G\ _{\infty} = 0$ against the shared scene constructor | [h13 preamble] |

The digest discipline is itself a measured correction: an earlier lane claimed “identical bytes” with no digest and a truncated 8-hex hash, and the rule now is that **every bit-identity claim carries the full digest** [HUNT, Modal].

**Uncontended timings.** All wall-clock figures quoted for the GPU path were rerun on a dedicated Modal L4 without contention, repeat spread 0.04–0.7 % [U20, summary line 1]. Two consequences: the  $N = 10\,000$  lattice ADMM/PGS ratio is **5.30×** uncontended (84.09 ms vs 445.61 ms) against 4.28× contended [U20 §1], and the previously published 3.47× at  $n_c = 256$  is **refuted** as a sweep-granularity artefact, the clean ratio being 0.38× [U20 §1, §3.1]. Modal billing for the rerun: 16.8 GPU-minutes over five jobs [U20 §0].

**Cross-device identity.** Five previously unmeasured suites were hashed on L4 (sm\_89), A10G (sm\_86) and H100 (sm\_90): **26 of 26 arrays byte-identical across all three** [U20 §2]; the 4 096-environment batch is identical across all four devices including the local RTX 5070 [U20, summary line 3].

**Where identity does not hold, and why.** Local-vs-cloud divergence was isolated to two causes in three subsystems [digests §0]:

| subsystem                     | diverges 5070 vs cloud | cause                                                         | the switch that proves it                                                                                                                       |
|-------------------------------|------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| lattice $N = 10\,000$         | yes                    | <b>device</b> — nvrtec floating-point contraction             | <code>fuse_fp=False</code> = identical 40-step state on both machines (3cddd8b0...); default splits at step 9, $\leq 6$ float32 ulp             |
| 1000:1 column                 | yes                    | <b>host</b> — NumPy BLAS/LAPACK on two x86 CPUs               | GPU ladder identical at all 13 iteration counts $\times 3$ codegen settings; first divergence is a host-built <code>Ja/Jb/bias</code> at step 3 |
| h1 cube, K8 tower             | yes                    | <b>host</b> — <code>np.linalg.eigvalsh</code> inside $\rho^*$ | $\rho$ differs on exactly the scenes that diverge; <code>fuse_fp=False</code> does not repair it                                                |
| npc_gpu cube, K8 tower 200 it | no                     | —                                                             | —                                                                                                                                               |

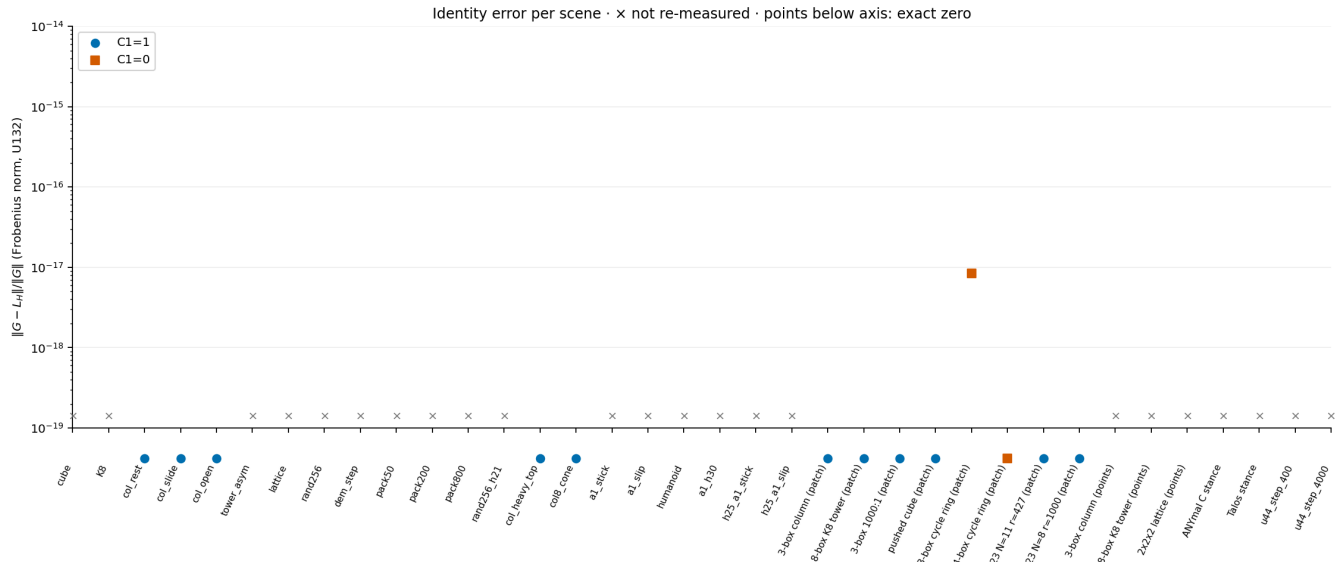
Three candidates were refuted in the process: a device-computed power-of-two int64 scale (bit-identical over a 200 484-point sweep), float32 vs float64 as the discriminator (float64 moves under contraction too), and launch-config-dependent reduction order (complete kernel inventory; every cross-thread accumulation is int64 or an integer counter, float reductions fixed at compile-time width  $NBLK = 256$ ) [digests §0]. `fuse_fp=False` is now the default of both solver modules, and the certificate says which side diverges [digests §8].

**Environment.** CPU lanes: OMP\_NUM\_THREADS=4 OPENBLAS\_NUM\_THREADS=4, NumPy 2.2.6, SciPy 1.15.3, Python 3.10.12, float64 throughout [U18 preamble; U21 “Metod”; U22 §1]. GPU lanes: Warp 1.13.0, local RTX 5070 12 GB (driver 580.178.04, sm\_120), cloud L4/A10G/H100 (driver 580.95.05) [U20 §0]. Shared-GPU contention is flagged wherever a local GPU millisecond is quoted [U22 §1; U33 preamble].

**[MISSING] items in this section:** uncontended (Modal) timings for the CPU router itself (U18), for the CPU Laplacian lanes (h11, h13, h15) and for U21/U22 — U20’s table covers ncp\_gpu, h1/h1b, h25, h30, U2b, U11, U8, U15 and the MPM suites, not these; and cross-device identity for those same CPU lanes.

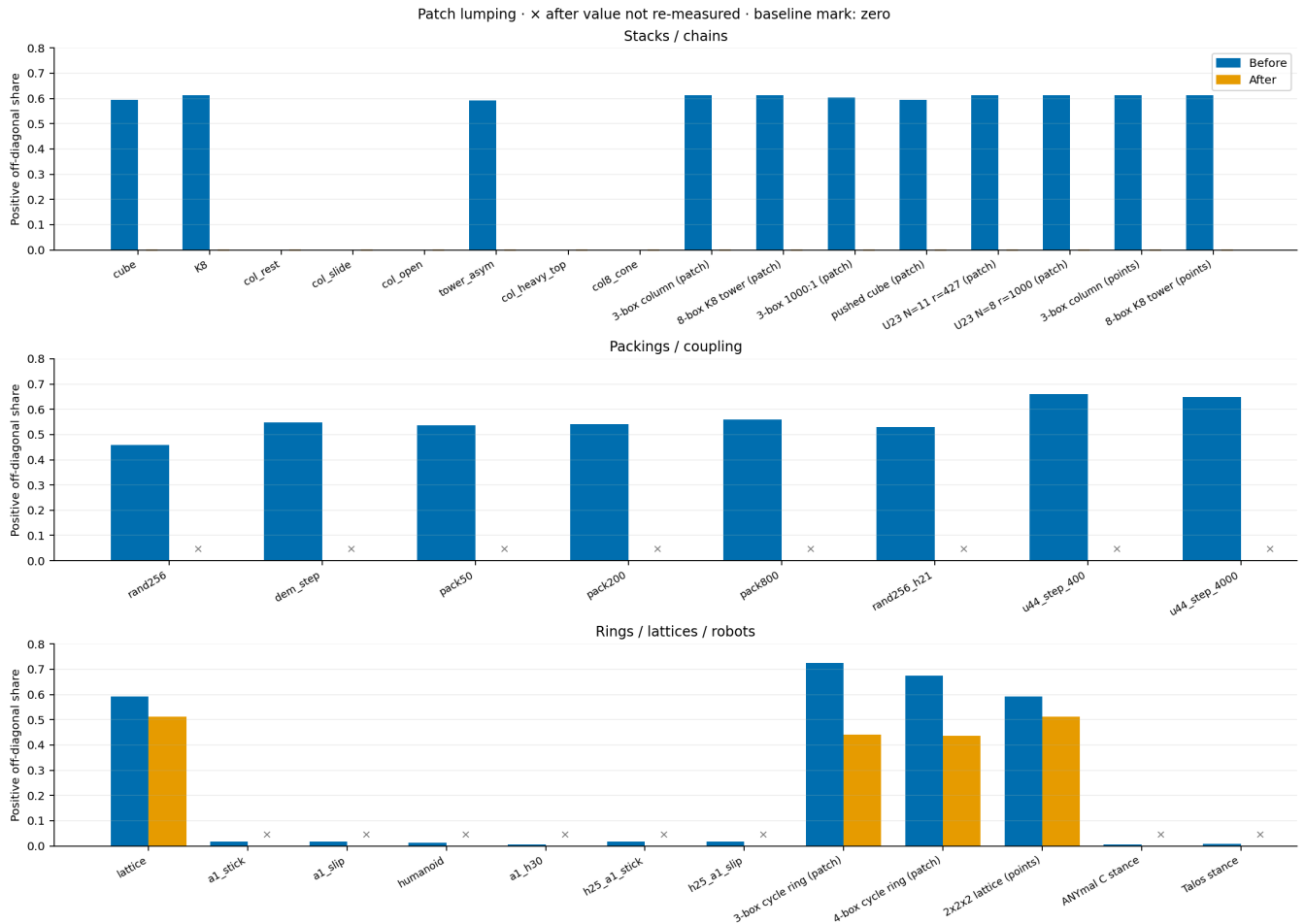
## Figures

All three figures read build/U132/p1\_figures\_data\_v2.csv (36 rows, sha256 213bb7c00482fa69b8d66aa741b1f2c41427cf803bc3e5ccbf6ac4c3a5bf798, identical over two measurement runs [U132 §“SHA-256 mellan körningarna”]). Each scene process builds J and Minv once and hashes both in its measurement JSON; A70’s saved J/Minv match byte for byte on its ten sampled scenes, and n\_c, grad\_max and C1 match U89 on all 36 rows [U132 §“Konstruktion och definitioner”]. Of A70’s 26 blocked fields, 16 now carry a measured value and 10 remain without a defined operator [U132 §1]. U132’s acceptance is RED on the completeness requirement: 25 computation checks pass, 35 delivery checks pass, 55 file pairs are byte-identical, and the gaps are left as gaps rather than filled with constants, factorisation residuals or a change of model [U132 §Acceptans, §“Kontroller, tider och misslyckanden”]. The plotted files are fig1\_v2.pdf–fig3\_v2.pdf, fig1\_data\_v2.csv–fig3\_data\_v2.csv and captions\_v2.md (sha256 1ff8d18919f024b8bb45ebd3016b9ed4bba2d9e4df2e22f1e4b8884f89ae16de; the full digest of every figure file is listed in [U132 §“SHA-256 mellan körningarna”]). U67’s CSV is not used.



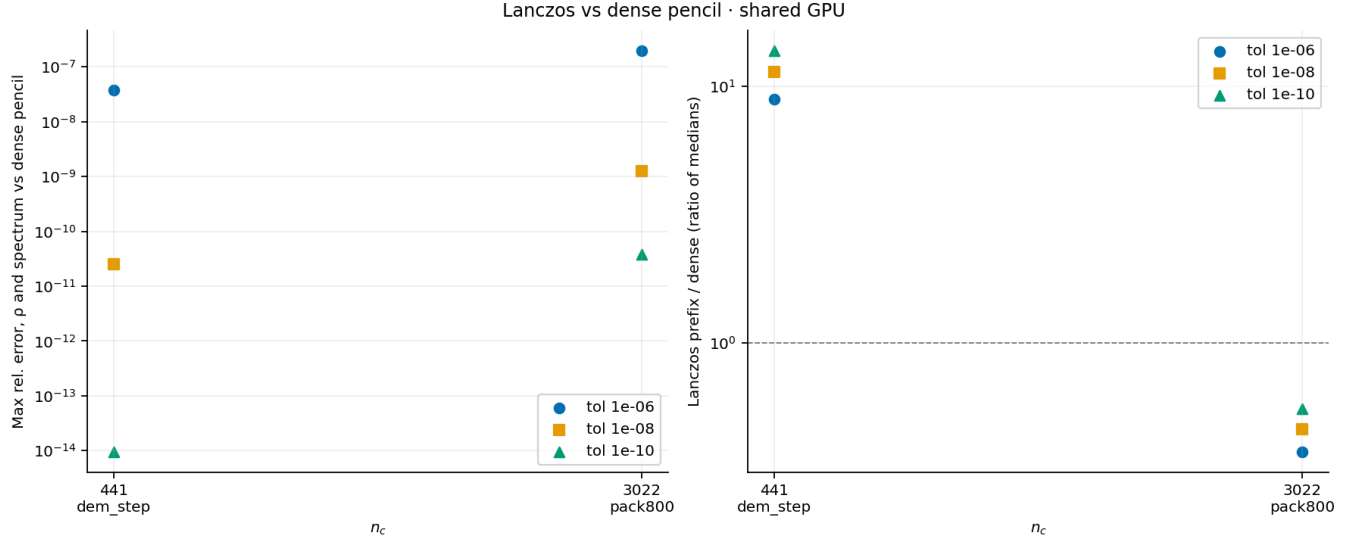
**Figure 1 — identity error against scene, with C1 status.** The identity error is  $\|G - L_H\|_F / \|G\|_F$ , with  $L_H$  built from the incidence and the body conductances in the common gauge; for point chains the construction is accepted only when the J blocks of each body recur with sign and Minv is block diagonal [U132 §“Konstruktion och definitioner”]. 13 rows carry a measured value: 12 read exactly 0 and 3-box cycle ring (patch) reads 8.351394258682832e-18; 11 of the 13 carry C1 = 1 and the two cycle rings carry C1 = 0, where the ground contact raises the body degree to 3 [U132 fig1\_data\_v2.csv; U89 §4]. The remaining 23 rows have no  $L_H$  for their J and every one of them carries C1 = 0 — **C1 falskt — identiteten är odefinierad**. No factorisation residual, no normal-block surrogate and no body-sum comparison is substituted for the missing operator; U67’s point check and the robot check each build a different surrogate, and their

diagnostics stay in the measurement JSON [U132 §“Konstruktion och definitioner”]. The figure draws those 23 rows as crosses and marks the 12 zeros below the log axis [U132 captions\_v2.md]. Four literals of U89 are withdrawn by the re-measurement: 3-box column (points), 8-box K8 tower (points) and 2x2x2 lattice (points) carried the constant 0.7500 and now carry no value, and col8\_cone carried 1.0000e−15 against a measured 0 [U132 §“A70:s 26 avvikelser”, §“Värdeändringar mot U89”]. The two degree-2 rings without ground contact are plotted from [U89 §4]: largest positive off-diagonal eigenvalue exactly 0.0000e+00 at  $n = 3, 4, 5, 6$ , against +189.2072 once the ground contact is added; the audit’s own ring reproduces  $\max \lambda^+ = 0.0$  on the off-diagonal blocks and records that the whole  $G$  is PSD, not NSD ( $\lambda_{\min} = -2.86907435568708e-15$ ,  $\lambda_{\max} = 48.3600364878805$ ) [A70 §3].



**Figure 2 — positive off-diagonal Frobenius share: the point model on all 36 rows, and the patch model on the 19 rows where a lumping operator is defined.** The share is  $\sqrt{\sum \max(\text{eig}(\text{sym}(G_{cd})), 0)^2} / \|G\|_F$  over  $c \neq d$ , with blocks under U89’s relative max-element bound omitted; A70’s loop over blocks reproduces the share inside the check tolerance [U132 §“Konstruktion och definitioner”]. Before-values are plotted on all 36 rows and agree with U89 to  $7.08e-06 \dots 4.00e-03$  relative [U132 §“Värdeändringar mot U89”]. Before-values on patch rows are the point model, while the structure fields and the after-values are the patch model: the columns do not describe the same operator [U132 §“Konstruktion och definitioner”; A70 §2]. After-values are plotted on 19 rows and are missing on 17, each [EJ OMMÄTT] with U132’s own status: the six packing rows (rand256, dem\_step, pack50, pack200, pack800, rand256\_h21) and the two STEP rows carry NO\_TRANSFORM\_DEFINED; before measured separately, and the eight articulated rows (a1\_stick, a1\_slip, humanoid, a1\_h30, h25\_a1\_stick, h25\_a1\_slip, ANYmal C stance, Talos stance) carry UNSPECIFIED\_ROBOT\_LUMPING; net-base-wrench candidate in diagnostics [U132 §“Luckor i hela CSV-filen”]. tower\_asym stays [EJ OMMÄTT] and is not plotted: its after-value 0 is computed on a tower with equal masses, not on the 1/1000 tower the row names, and A70 §7 restricts that field

from use as a scene-matched lumping measurement (ONLY\_AS\_STRUCTURAL\_NSD\_EXAMPLE) [A70 §7, field\_restrictions.csv]. The robot candidate U132 built and did not carry into the column is  $C = J[:, :6]^T$ , the net wrench on the base coordinates: rank  $J = 12$  against rank  $C = 6$  on the six quadruped rows, virtual-work loss  $\|J^T - J_w^T C\|_F / \|J\|_F = 0.24550817363113317 \dots 0.29935972922742543$ , against a singular-value floor of  $0.21291003099264055 \dots 0.2826634346470032$  for any operator of rank at most 6; on humanoid and Talos stance rank  $C = 5$  is below rank  $J = 6$  [U132 §“Robotlumpning”]. One block remains after that reduction, so its zero share is a property of the construction and not a measurement of conservation [U132 §“Robotlumpning”].



**Figure 3 — pencil spectrum: accuracy and cost against a dense eigensolve.** The figure reproduces U89’s Lanczos prefix and error for dem\_step and pack800; U132 measured no new timings, and **GPU shared** carries over to the source timings [U132 captions\_v2.md, §“Kontroller, tider och misslyckanden”]. The error metric is the max relative error in  $\rho$ ,  $\lambda_{\min}^+$  and  $\lambda_{\max}$  against the dense pencil; the prefix stops at the last Lanczos iteration and excludes the reduced eigensolve and the result extraction; the ratio divides the medians [U132 fig3\_data\_v2.csv, captions\_v2.md]. At  $n_c = 3\,022$ , 140 / 160 / 180 steps: dense median 2 336.51 ms against Lanczos prefix medians 874.06 / 1 072.23 / 1 294.51 ms at tol 1e−6 / 1e−8 / 1e−10, ratio Lanczos/dense 0.3741 / 0.4589 / 0.5540, max relative error 1.9914e−07 / 1.2467e−09 / 3.7924e−11, 100/100 inside tolerance. At  $n_c = 441$ , 100 / 120 / 140 steps: dense median 59.07 ms against 523.83 / 669.96 / 813.21 ms, ratio 8.8680 / 11.3418 / 13.7669, max relative error 3.8170e−08 / 2.5240e−11 / 9.3780e−15, 100/100 [U132 fig3\_data\_v2.csv]. The audit’s interleaved pairs are a separate measurement under a different load: dense median 1 673.143 ms, Lanczos full 611.307 ms, prefix 598.212 ms, pairwise ratios in [1.252073, 3.535732], 100/100; the ratio of the medians is 2.736994 and the median of the pairwise ratios is 2.691977, and these are different summary measures [A70 §5]. Accuracy at  $n_c = 3\,022$  against the dense pencil: mean relative error 1.64358963e−16, maximum 1.37149392e−14 [A63 §“U67: pencil och kondition”].

**Relation between the figures and table 6.** The three figures still read U132’s figure CSV and do not plot  $\rho$  predictions. Table 6 instead reads U135’s independent 25-point sweep over 21 scenes; it reports grid minima, tie-breaking toward the lowest  $\rho$ , censored points and numerical failures [U135 §§Tabell, Censurering, numerikfel och delade minima]. The historical oracle column of U83 remains withdrawn [A70 §6, §7].

## Appendix A — consolidated headline table

| row                                              | number                                                                                                                                   | source                     |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| patch Delassus = graph-assembled block Laplacian | 9.586e−17 relative (K8 tower)                                                                                                            | [h13 §1B]                  |
| off-diagonal blocks negative semidefinite        | $\max \lambda^+ = -6.259\text{e}-04$ (K8)                                                                                                | [h13 §1B]                  |
| block row sum = graph Dirichlet residue          | $\leq 4.8\text{e}-17$ on the three scenes that pass; $1.9\text{e}-16$ including the $2\times 2\times 2$ lattice, which does not          | [h13 §1B]                  |
| friction does not break the scalar identity      | $G = L \otimes I_3$ , cross blocks exactly 0                                                                                             | [08 §1]                    |
| breaks at body degree $\geq 3$                   | $\max \lambda^+ = +6.357\text{e}-01$ ( $2\times 2\times 2$ lattice)                                                                      | [h13 §1E]                  |
| $\text{spec}(G)\{0\} = \text{spec}(L_v, M)$      | $4.8\text{e}-16 \dots 1.1\text{e}-15$ relative                                                                                           | [h13 §1G]                  |
| spectrum estimated matrix-free                   | $\leq 204$ matvecs; $\lambda_{\min}^+$ rel err $1.2\text{e}-15 \dots 1.8\text{e}-9$ , $\lambda_{\max}$ rel err $0.0 \dots 2.1\text{e}-7$ | [h15 §1A]                  |
| estimator cost share (GPU)                       | 2.5 % at $n_c = 4\,096$                                                                                                                  | [h1 §H1b.1]                |
| 1000:1 column, cone binds                        | 18 225 PGS $\rightarrow$ 216 ADMM at $\rho = \rho^*/10$ ; 9 215 $\rightarrow$ 29 at $\rho = \lambda_{\min}^+$ with the set binding       | [h11 §1A, §1C; h15 §2]     |
| K8 tower, patch                                  | 14 630 PGS $\rightarrow$ 140 ADMM ( $105\times$ )                                                                                        | [h13 §2]                   |
| K8 tower, point                                  | 14 682 PGS $\rightarrow$ 8 ( $\rho^*/10$ ) $\rightarrow$ 3 ( $\lambda_{\min}^+$ )                                                        | [h11 §1A; h15 §2]          |
| asymmetric 1/1000 tower                          | 1 512 $\rightarrow$ 9 at $\rho = \lambda_{\min}^+$ ( $168\times$ )                                                                       | [h15 §2]                   |
| chain x-update                                   | block tridiagonal, $1.67\text{e}-16$ vs dense Cholesky                                                                                   | [h13 §2D]                  |
| patch rank / derivative                          | rank J 6/6 vs 6/12; cond 72.6 vs $4.9\text{e}16$ ; $dw/d\mu$ vs FD $1.9\text{e}-10$                                                      | [h13 §3]                   |
| router over 21 scenes                            | 52.4 s total; PGS-100 35.8 s but converges on 8/21; router $\rho$ is $1.6\text{--}3.1\times$ from U135's swept optimum where used        | [A78 §§2, 4; U135 §Tabell] |
| router after the A1 branch                       | 1/21 (articulated) or 0/21 ( $n_c \leq 4$ )                                                                                              | [U33 §1]                   |
| $\lambda_{\min}^+$ as a stability margin         | constant $2.3331\text{e}-03$ to toppling, $r = 0.0000$                                                                                   | [U22 §2.1]                 |
| certificate margin vs ZMP / Coulomb              | coincident K8 zero crossings; max pointwise discrepancy 0.97 on 40 randomized towers                                                     | [U22 §2.1, §5; A81]        |
| effective resistance vs degree                   | Spearman $+0.401$ vs $+0.339$                                                                                                            | [U21 §1]                   |
| grain-scale law                                  | $\lambda_{\min} \propto d^{1.262}$ , $F^* \propto d^{1.133}$ (U7 ref 1.22 [1.07, 1.38]); $n = 4$ points, no CI                           | [U21 §2]                   |
| uncontended ADMM/PGS at 40 000 contacts          | $5.30\times$                                                                                                                             | [U20 §1]                   |
| cross-device identity                            | 26/26 arrays over L4/A10G/H100                                                                                                           | [U20 §2]                   |

## Appendix B — what is not claimed

- The proof is Appendix C. No claim in Appendix C is measured by Appendix C; the numbers cited next to each step were produced by the lanes named, and none of those lanes carries an independent audit (§6.8).
- The bibliography is the References section: 36 entries checked against the publisher of record, of which 1 (Thomas 1949) could not be found and is marked **EJ FUNNEN**, plus entry 37 added for §6.9. Works named in §6.9 without a verified entry — Spielman-Teng, Koutis-Miller-Peng, Kyng-Sachdeva — carry no entry. [MISSING: verified entries for the nearly-linear Laplacian solvers]

- No comparison against a published external solver’s iteration counts on these scenes is included here; the external oracles that exist in the corpus (Box3D, MuJoCo, Pinocchio, coal) were run for other claims and are not part of this paper’s tables. No run of the Acary-Cadoux-Lemaréchal-Malick formulation [5] or of Cadoux’ algorithm [6, 7] on these scenes is included either; the outer loop named after Cadoux is this paper’s own implementation. §6.9 lists the named comparisons that were not run. [MISSING: baseline runs of [5], [6], [7]]
- The Contensou/spin part of the patch cone is a model change; the measured gap ( $2.5e-05 \dots 3.4e-05$  on these scenes) is not evidence of general agreement [h13 §2B], and the spin counterexample measures 0.3232 and 0.2767 relative wrench error — the  $\sqrt{3}$  capacity ratio between four corner cones and the patch cone [U89 §2; A70 §4]. Lumping restores rank and the off-diagonal sign structure; it does not preserve the contact solution under spin.
- No spectral  $\rho$  rule is claimed for a mixed stick/slip active set, and no impossibility result is claimed either: the measured statement is that the iteration map has no Fréchet Jacobian at those states [U95 §5; A68 §5].
- The one-contact theorem  $\rho_{\text{opt}} = g/(1+\mu^2)$  is a statement about the spectral radius of the map with the shift refreshed every iteration. It is not a formula for the iteration count from a cold start, and not a rule for  $n_c > 1$  [A68 §§1, 2, 6].
- Nothing here says which of  $\rho = \lambda_{\min}^+$  and  $\rho = \rho^*$  a solver should pick on an unseen packing;  $\text{cond}(G)$  does not separate them [HUNT, H1b].

## Appendix C — Proof

This appendix discharges the [MISSING] of §2.3 and the first bullet of Appendix B: it is the algebraic proof of the theorem of §2, its conditions C1–C6, and the three corollaries, with every hypothesis stated and every step justified. Nothing here is new physics; what is new relative to the source files is that the statements are *proved* rather than assembled from measured columns. Where a step has a measured counterpart in the corpus, the measurement is cited next to the step — the two are independent, and they agree.

### C.0 Notation and standing hypotheses

**(N1) Bodies.** A finite set of bodies  $k = 1 \dots n_b$ , each with generalized velocity  $v_k \in \mathbb{R}^{d_k}$  and symmetric positive definite inertia  $M_k \in \mathbb{R}^{d_k \times d_k}$ . Two cases are carried in parallel throughout:

- *point masses*:  $d_k = 3$ ,  $M_k = m_k I_3$ ,  $M_k^{-1} = (1/m_k) I_3$ ;
- *rigid bodies*:  $d_k = 6$ ,  $M_k = \text{blkdiag}(m_k I_3, I_k)$  in a body-centred frame,  $I_k > 0$ .

Write  $v = (v_1, \dots, v_{n_b}) \in \mathbb{R}^{n_v}$ ,  $n_v = \sum_k d_k$ ,  $M = \text{blkdiag}(M_1, \dots, M_{n_b}) > 0$ , and let  $P_k : \mathbb{R}^{n_v} \rightarrow \mathbb{R}^{d_k}$  be the coordinate projection onto body  $k$ ’s block, so that  $P_k M^{-1} P_k^T = \delta_{\{kl\}} M_k^{-1}$ . Static (ground) bodies are *not* members of the list: they carry  $M^{-1} = 0$  and are handled in (N2) by an incidence of zero. This is the convention of `h13_patch.py:incidence` (“over DYNAMIC bodies only; the static ground has zero conductance and drops out”) and of 08 §1 (“the static ground column dropped”).

**(N2) Contacts and incidence.** A finite set of contacts  $c = 1 \dots n_c$ . Each contact is an **ordered** pair  $(a_c, b_c)$  of distinct bodies, at most one of which may be static. The **incidence sign** is

$$\begin{aligned} \sigma_{ck} &= +1 && \text{if } k = a_c \\ &= -1 && \text{if } k = b_c \\ &= 0 && \text{otherwise} \end{aligned} \quad (k \text{ static} \Rightarrow \sigma_{ck} := 0 \text{ as well})$$

Collect the incidence column of body  $k$  as  $s_k = (\sigma_{\{1k\}}, \dots, \sigma_{\{n_c k\}})^T \in \mathbb{R}^{n_c}$ , and write  $\text{deg}(k) = \#\{c : \sigma_{ck} \neq 0\}$  for the **body degree** — the number of contacts touching body  $k$ . Two

contacts  $c \neq c'$  are *adjacent* if they share a body.

**(N3) Contact Jacobian.** Contact  $c$  measures  $m_c$  scalar relative velocities,  $m_c \in \{1, 3, 6\}$  (normal only / normal+two tangents / a 6-dimensional patch twist). Its row block is

$$J_c = \sum_k \sigma_{ck} B_{ck} P_k \in R^{\{m_c \times n_v\}}, \quad J = (J_1; \dots; J_{\{n_c\}}) \in R^{\{(\sum m_c) \times n_v\}}$$

where  $B_{ck} \in R^{\{m_c \times d_k\}}$  is the **kinematic block** of contact  $c$  at body  $k$ : the map from body  $k$ 's generalized velocity to the  $m_c$  measured directions, carrying no sign of its own. For a point contact at  $p$  on a rigid body with centre  $c_k$  and frame  $E_c = [n_c; t1_c; t2_c]$  (rows orthonormal),  $B_{ck} = [E_c, -E_c \text{ skew}(p - c_k)]$ ; for a point mass,  $B_{ck} = E_c$ ; for a patch,  $B_{ck}$  is the  $6 \times 6$  block of `h13_patch.py:_patch_rows_local` divided by its sign. The Delassus operator is

$$G = J M^{-1} J^T, \quad G_{\{cc'\}} = J_c M^{-1} J_{\{c'\}}^T \in R^{\{m_c \times m_{\{c'\}}\}}.$$

**(N4) The common gauge.** Fix one world point  $p_{\text{ref}}$  and world axes. For a rigid body let  $Ad_k = Ad(p_{\text{ref}} - c_k) = [[I_3, -\text{skew}(p_{\text{ref}} - c_k)], [0, I_3]]$ , the twist-transport map to the body-fixed material point momentarily at  $p_{\text{ref}}$  (`h13_patch.py:adjoint`). The **common gauge** is the choice  $B_{ck} = Ad_k$  for every contact  $c$  incident to  $k$  — i.e. all contacts measure the same relative twist, at the same point, in the same axes (`h13_patch.py:J_common`). Define the **conductance**

$$W_k := Ad_k M_k^{-1} Ad_k^T \quad (6 \times 6 \text{ for rigid bodies, } (1/m_k) I_3 \text{ for point masses}).$$

**Lemma C.0.**  $W_k$  is symmetric positive definite for every dynamic body, and  $W_k = 0$  for static ones. *Proof.*  $Ad_k$  is block unit-triangular, hence invertible with  $\det Ad_k = 1$ ;  $M_k^{-1} > 0$  because  $m_k > 0$  and  $I_k > 0$ . A congruence by an invertible matrix preserves positive definiteness:  $x^T W_k x = (Ad_k^T x)^T M_k^{-1} (Ad_k^T x) > 0$  for  $x \neq 0$  since  $Ad_k^T x \neq 0$ . Symmetry is immediate from  $M_k^{-1} = M_k^{-T}$ . For a static body  $M_k^{-1} = 0$ . ■

(Measured counterpart:  $\min \text{eig } W_i = 7.99e-1 / 3.07e-1 / 7.99e-4 / 7.87e-1$  and  $\|W_i - W_i^T\| = 0.000e+00$  on the four scenes, [h13 §1C].)

**(N5) The conditions of §2.2, restated as hypotheses.**

- **(C1)** *One contact per body pair*: the map  $c \mapsto \{a_c, b_c\}$  is injective.
- **(C2)**  $\text{conductance} = W_k$  — this is a *definition* under (N4), not a hypothesis; it is the content of Lemma C.1 below that  $W_k$  is what appears.
- **(C3)** normal angle  $n_c \cdot n_{\{c'\}} \geq 0$  — used only in the scalar specialisation, Corollary C.4.
- **(C4)** translation-only bodies — used only in the scalar specialisation.
- **(C5)** friction rows — Proposition C.5.
- **(C6)**  $\deg(k) \leq 2$  for every dynamic body — Theorem C.2.

## C.1 The assembly lemma (statement (i))

**Lemma C.1 (assembly).** Under (N1)–(N3), for all contacts  $c, c'$  (equal or not)

$$G_{\{cc'\}} = \sum_k \sigma_{ck} \sigma_{\{c'k\}} B_{ck} M_k^{-1} B_{\{c'k\}}^T,$$

the sum running over the bodies incident to **both**  $c$  and  $c'$ ; in particular  $G_{\{cc'\}} = 0$  when  $c$  and  $c'$  share no body. If (C1) holds and  $c \neq c'$ , the sum has at most one term: if  $c$  and  $c'$  share the body  $k$ ,

$$G_{\{cc'\}} = \sigma_{ck} \sigma_{\{c'k\}} \cdot B_{ck} M_k^{-1} B_{\{c'k\}}^T,$$

and in the common gauge (N4) this is  $\sigma_{ck} \sigma_{\{c'k\}} W_k$ . The diagonal is  $G_{\{cc\}} = \sum_k \sigma_{ck}^2 B_{ck} M_k^{-1} B_{ck}^T = \sum_{\{k \in \{a_c, b_c\}\}} B_{ck} M_k^{-1} B_{ck}^T$ , which in the common gauge is  $W_{\{a_c\}} + W_{\{b_c\}}$  (a static member contributing 0).

*Proof.* Expand using (N3) and the block-diagonality of  $M^{-1}$ :

$$\begin{aligned} G_{\{cc'\}} &= J_c M^{-1} J_{c'}^T \\ &= \left( \sum_k \sigma_{ck} B_{ck} P_k \right) M^{-1} \left( \sum_l \sigma_{\{c'l\}} B_{\{c'l\}} P_l \right)^T \\ &= \sum_k \sum_l \sigma_{ck} \sigma_{\{c'l\}} B_{ck} \left( P_k M^{-1} P_l^T \right) B_{\{c'l\}}^T \\ &= \sum_k \sum_l \sigma_{ck} \sigma_{\{c'l\}} B_{ck} \left( \delta_{\{kl\}} M_k^{-1} \right) B_{\{c'l\}}^T \\ &= \sum_k \sigma_{ck} \sigma_{\{c'k\}} B_{ck} M_k^{-1} B_{\{c'k\}}^T, \end{aligned}$$

the third equality being (N1)'s  $P_k M^{-1} P_l^T = \delta_{\{kl\}} M_k^{-1}$ . A term survives only when  $\sigma_{ck} \sigma_{\{c'k\}} \neq 0$ , i.e. only for bodies incident to both contacts; static bodies contribute  $M_k^{-1} = 0$  (equivalently  $\sigma_{ck} = 0$  by (N2)). Hence the stated support.

For the “at most one term” claim, suppose  $c \neq c'$  share two distinct bodies  $k \neq l$ . Then both  $k$  and  $l$  are incident to  $c$  and to  $c'$ , and since each contact is incident to exactly two bodies,  $\{a_c, b_c\} = \{k, l\} = \{a_{c'}, b_{c'}\}$ : two distinct contacts on the same body pair, contradicting (C1). So under (C1) distinct contacts share at most one body. The diagonal statement is the case  $c = c'$  with  $\sigma_{ck}^2 = 1$  on the (at most two) incident bodies. Substituting  $B_{ck} = Ad_k$  from (N4) gives  $B_{ck} M_k^{-1} B_{\{c'k\}}^T = Ad_k M_k^{-1} Ad_k^T = W_k$ , independent of which contacts  $c, c'$  are involved. ■

**Corollary C.1.1 (Kronecker form and gauge covariance — statement (ii), first half).** *In the common gauge with all  $m_c = 6$  (patch lumping, so that every block has the same size),*

$$G_{\text{common}} = \sum_k (s_k s_k^T) \otimes W_k.$$

*If each contact is re-expressed through an invertible per-contact gauge  $D_c \in R^{6 \times 6}$ , i.e.  $J_c^{\text{loc}} = D_c J_c^{\text{common}}$ , then with  $D = \text{blkdiag}(D_1, \dots, D_{n_c})$*

$$G_{\text{loc}} = D G_{\text{common}} D^T = D \left( \sum_k (s_k s_k^T) \otimes W_k \right) D^T,$$

*which is the form  $G = B \left( \sum_k \sigma \sigma^T \otimes W_k \right) B^T$  of §2 with  $B = D$ .*

*Proof.* Block  $(c, c')$  of  $\sum_k (s_k s_k^T) \otimes W_k$  is  $\sum_k \sigma_{ck} \sigma_{\{c'k\}} W_k$ , which is  $G_{\{cc'\}}$  by Lemma C.1 in the common gauge — note that no hypothesis (C1) is needed for this identity, only (N4). The second statement is  $J^{\text{loc}} = D J^{\text{common}}$ , hence  $G_{\text{loc}} = J^{\text{loc}} M^{-1} J^{\text{loc}^T} = D J^{\text{common}} M^{-1} J^{\text{common}^T} D^T = D G_{\text{common}} D^T$ . ■

The patch gauge of the corpus is  $D_c = S_c \cdot \text{PERM} \cdot \text{blkdiag}(E_c, E_c) \cdot \text{Ad}(p_c - p_{\text{ref}})$  (h13\_patch.py:patch\_gauge), invertible because  $S_c = \text{diag}(1, e_2, e_1, 1, 1, l)$  has positive diagonal whenever the patch has positive extent, PERM is a permutation,  $\text{blkdiag}(E_c, E_c)$  is orthogonal and  $\text{Ad}(\cdot)$  is unit-triangular.  $G_{\text{loc}} = D G_{\text{common}} D^T$  is a **congruence** and not a similarity, so  $\text{spec}(G_{\text{loc}}) \neq \text{spec}(G_{\text{common}})$  in general, and the off-diagonal sign structure is **not** preserved: with  $W_k = I_3$ ,  $D_c = I_3$  and  $D_{\{c'\}} = \text{diag}(-1, -1, 1)$  — a rotation by  $\pi$  about the normal, an admissible per-contact gauge — the block  $-W_k \leq 0$  maps to  $-D_c W_k D_{\{c'\}}^T = \text{diag}(1, 1, -1)$ , which is symmetric with two positive eigenvalues. The Laplacian verdict of Theorem C.2 is therefore defined in the common gauge only; the local gauge is used for the solve, where the cone is the normalised one. Measured:  $\|G_{\text{local}} - D G_{\text{common}} D^T\|/\|G\| \leq 3.197\text{e-}15$  [h13 §1F], and the local-gauge spectra differ from the common-gauge ones by up to  $26\times$  in condition number [h15 §1F].

## C.2 The Laplacian characterisation (statement (ii))

**Definition C.2.0 (the line graph and its block Laplacian).** Let  $H$  be the graph with vertex set the contacts  $\{1, \dots, n_c\}$ , together with, for every dynamic body  $k$ ,

- an **edge**  $\{c, c'\}$  of block weight  $W_k$  if  $\deg(k) = 2$  with incident contacts  $c \neq c'$ ;
- a **Dirichlet (ground) half-edge** at  $c$  of block weight  $W_k$  if  $\deg(k) = 1$  with incident contact  $c$ ;
- nothing if  $\deg(k) = 0$ .

Its grounded weighted block Laplacian is

$$L_H = \sum_{\{k : \deg(k)=2\}} (e_c - e_{\{c'\}})(e_c - e_{\{c'\}})^T \otimes W_k + \sum_{\{k : \deg(k)=1\}} e_c e_c^T \otimes W_k.$$

Under condition (3) of Theorem C.2,  $H$  is the **line graph** of the contact graph (bodies  $\leftrightarrow$  vertices, contacts  $\leftrightarrow$  edges), with the pendant half-edges of degree-1 bodies as the Dirichlet terms. At a body of degree  $d \geq 3$  the line graph carries a  $K_d$  clique of weight  $W_k$ , which Definition C.2.0 does not cover; that case is the signed assembly of the last paragraph of Theorem C.2.

**Theorem C.2 (characterisation).** *Assume (N1)–(N4) and (C1), all blocks of size 6 (patch lumping) or all of size 1 (scalar normal-only), read in the common gauge. Then the following are equivalent:*

1. every off-diagonal block  $G_{\{cc'\}}$ ,  $c \neq c'$ , is negative semidefinite;
2.  $\sigma_{ck} \sigma_{\{c'k\}} = -1$  for every dynamic body  $k$  and every pair of distinct contacts  $c, c'$  at  $k$ ;
3. every dynamic body has  $\deg(k) \leq 2$ , and at every body with  $\deg(k) = 2$  the two incidence signs are opposite.

Under these equivalent conditions

$$G = L_H$$

exactly —  $G$  is a grounded weighted block Laplacian on the line graph of the contact graph, with edge conductance the shared body's inverse (spatial) inertia — and the block row sums satisfy

$$\sum_{\{c'\}} G_{\{cc'\}} = \sum_k \sigma_{ck} (\sum_{\{c'\}} \sigma_{\{c'k\}}) W_k,$$

which vanishes on every body of degree 2 and leaves exactly the Dirichlet residue of the degree-1 bodies. Without the conditions,  $G = \sum_k (s_k s_k^T) \otimes W_k$  still holds and  $G \geq 0$  still holds. Under (C1) with some dynamic body of degree  $\geq 3$ , some off-diagonal block equals  $+W_k > 0$ . Without (C1) an off-diagonal block is a sum  $\sum_k \sigma_{ck} \sigma_{\{c'k\}} W_k$  over the shared bodies and may be positive definite, indefinite or negative semidefinite; the measured case  $+(1/m_a + 1/m_b)$  of two same-sign point contacts at one interface is the positive one [h13 §1A].

*Proof.*

**(2)  $\Rightarrow$  (1).** Let  $c \neq c'$ . By (C1) and Lemma C.1, either they share no body and  $G_{\{cc'\}} = 0 \leq 0$ , or they share exactly one body  $k$  and  $G_{\{cc'\}} = \sigma_{ck} \sigma_{\{c'k\}} W_k = -W_k$ , which is negative definite, hence negative semidefinite, by Lemma C.0.

**(1)  $\Rightarrow$  (2).** Suppose (2) fails: there are a dynamic body  $k$  and contacts  $c \neq c'$  at  $k$  with  $\sigma_{ck} \sigma_{\{c'k\}} = +1$ . By (C1) and Lemma C.1 the shared body is the *only* term, so  $G_{\{cc'\}} = +W_k$ , which is symmetric with  $\lambda_{\min}(W_k) > 0$  (Lemma C.0); its symmetric part is itself and has a strictly positive eigenvalue, so it is not negative semidefinite, contradicting (1).

**(2)  $\Rightarrow$  (3).** Suppose some dynamic body  $k$  has  $\deg(k) = d \geq 3$ , with incident contacts  $c_1, \dots, c_d$  and signs  $\sigma_{\{c_1 k\}}, \dots, \sigma_{\{c_d k\}} \in \{-1, +1\}$ . Three values drawn from a two-element set contain, by the pigeonhole principle, a repeated value: there are  $i \neq j$  with  $\sigma_{\{c_i k\}} = \sigma_{\{c_j k\}}$ , hence  $\sigma_{\{c_i k\}} \sigma_{\{c_j k\}} = +1$ , contradicting (2). So  $\deg(k) \leq 2$ ; and at a degree-2 body (2) applied to its single pair says the two signs are opposite.

**(3)  $\Rightarrow$  (2).** At a body of degree  $\leq 1$  there is no pair and the condition is vacuous; at a body of degree 2 the only pair is the one (3) constrains.

**The assembly  $G = L_H$ .** By Corollary C.1.1,  $G = \sum_k (s_k s_k^\top) \otimes W_k$  in the common gauge. Under (3): if  $\deg(k) = 2$  with contacts  $c, c'$  of opposite sign then  $s_k = \pm(e_c - e_{c'})$ , and since  $(-s)(-s)^\top = s s^\top$  the overall sign is immaterial, so  $s_k s_k^\top = (e_c - e_{c'})(e_c - e_{c'})^\top$ ; if  $\deg(k) = 1$  with contact  $c$  then  $s_k = \pm e_c$  and  $s_k s_k^\top = e_c e_c^\top$ ; if  $\deg(k) = 0$  then  $s_k = 0$ . Summing term by term reproduces Definition C.2.0 exactly, so  $G = L_H$ . Note where each hypothesis is used: (C1) is needed for the *characterisation*  $(1) \Leftrightarrow (2)$  — without it an off-diagonal block would be a sum  $\sum_k \sigma_{ck} \sigma_{c'k} W_k$  over several shared bodies whose sign is not determined term by term, which is precisely the  $+(1/m_a + 1/m_b)$  off-diagonal of two same-sign point contacts at one interface [08 §1; h13 §1A, max off-diagonal  $+2.0000e+0$  on four scenes] — while (3) is needed for the *identification* with  $L_H$ .

**Row sums.** Summing Lemma C.1 over  $c'$  (including  $c' = c$ ):  $\sum_{c'} G_{\{cc'\}} = \sum_{c'} \sum_k \sigma_{ck} \sigma_{c'k} W_k = \sum_k \sigma_{ck} (\sum_{c'} \sigma_{c'k}) W_k$ , the exchange of finite sums being legitimate. If  $\deg(k) = 2$  with opposite signs then  $\sum_{c'} \sigma_{c'k} = 0$  and the body drops out; if  $\deg(k) = 1$  at contact  $c_0$  then it contributes  $\sigma_{ck} \sigma_{c_0k} W_k$ , i.e.  $W_k$  when  $c = c_0$  and 0 otherwise. The surviving sum is the grounded/Dirichlet term of Definition C.2.0. (Measured: block row sums equal the graph's Dirichlet residue to  $\leq 1.9e-16$  [h13 §1B, column 5].)

**Positive semidefiniteness without the conditions.**  $G = J M^{-1} J^\top = (J M^{-1/2})(J M^{-1/2})^\top \geq 0$  for any  $J$ , since  $M > 0$  has a symmetric positive definite square root. The assembly  $G = \sum_k (s_k s_k^\top) \otimes W_k$  needed only (N4), so it survives degree  $\geq 3$  as well: what fails is the sign condition, not the graph assembly. ■

**Corollary C.2.1 (the degree-3 obstruction is orientation-free — C6).** *No relabelling of the contacts'  $(a_c, b_c)$  order — equivalently, no choice of signs  $\theta_c \in \{-1, +1\}$  replacing  $J_c$  by  $\theta_c J_c$  — can make all off-diagonal blocks negative semidefinite if some dynamic body has degree  $\geq 3$ . If every dynamic body has degree  $\leq 2$ , such a choice always exists, whatever the parity of any cycle in the contact graph.*

*Proof.* Switching by  $\theta$  sends  $\sigma_{ck} \mapsto \theta_c \sigma_{ck}$ , hence  $\sigma_{ck} \sigma_{c'k} \mapsto \theta_c \theta_{c'} \sigma_{ck} \sigma_{c'k}$ ; it is a congruence of  $G$  by the signature matrix  $\text{blkdiag}(\theta_c I)$ , so it changes no spectrum. The pigeonhole argument of  $(2) \Rightarrow (3)$  is invariant under it: at a body of degree  $d \geq 3$  the  $d$  switched signs  $\theta_{c_i} \sigma_{c_i k}$  still lie in  $\{\pm 1\}$ , so two of them still coincide. Hence degree  $\geq 3$  is an obstruction for every orientation.

Conversely, let every dynamic body have degree  $\leq 2$ . Delete every static vertex from the contact graph: by (N2) a static body carries  $\sigma_{ck} = 0$ , so it contributes no term to any  $G_{\{cc'\}}$  and no edge or half-edge to  $H$ . In the remaining graph every vertex is dynamic and has degree  $\leq 2$ , so each connected component is a path or a cycle, and a contact whose other endpoint was static becomes a pendant edge of that component. Define balance of the signed graph  $(H, -\varepsilon)$  in the sense of Harary, for which it suffices that every cycle of  $H$  carry  $\Pi(-\varepsilon) = +1$ . On a path component  $H$  is a path, which is acyclic, and a switching always exists (propagate  $\theta$  along the path). On a cycle component with bodies  $b_1, \dots, b_L$  and contacts  $c_i = (b_i, b_{i+1})$ , indices mod  $L$ , the line graph is the  $L$ -cycle  $c_1 \dots c_L$  and

$$\begin{aligned} \prod_{i=1}^L \varepsilon_{\{c_i, c_{i+1}\}} &= \prod_{i=1}^L \sigma_{\{c_i, b_{i+1}\}} \sigma_{\{c_{i+1}, b_{i+1}\}} \\ &= \prod_{i=1}^L \sigma_{\{c_i, b_i\}} \sigma_{\{c_i, b_{i+1}\}} \\ &= \prod_{i=1}^L (+1)(-1) = (-1)^L, \end{aligned}$$

the second equality regrouping the  $2L$  factors by contact (each contact  $c_i$  occurs in exactly two edges of the cycle, contributing  $\sigma_{\{c_i, b_{i+1}\}}$  and  $\sigma_{\{c_i, b_i\}}$ ), and the third using that a contact has one  $+1$  and one  $-1$  incidence by (N2). Hence  $\Pi(-\varepsilon) = (-1)^L (-1)^L = +1$  for every  $L$ , the signed graph is balanced, and a switching to  $\varepsilon \equiv -1$  exists. ■

*Remark (physical admissibility, and a refinement of h13's parenthetical).* Switching  $\theta_c = -1$  sends  $\lambda_c \mapsto -\lambda_c$  and therefore  $K_\mu \mapsto -K_\mu$ : it is legitimate for the *matrix* statement, which is what

the Laplacian verdict is about, but must be undone before the complementarity problem is solved. With the physical convention fixed (normal pointing from  $b\_c$  to  $a\_c$ , “upper body is a”), a stack or chain already satisfies (3) with no switching at all: the middle body of a stack is the a-side of the patch below it and the b-side of the patch above it, so its two incidence signs are +1 and -1. h13 §1E records the degree- $\leq 2$  case as “columns, towers, chains, and even cycles”; the computation above shows the parity is immaterial — a cycle of **any** length admits the orientation, since the cyclic orientation makes every body one contact’s a-side and the other’s b-side. What the corpus measures is the +1 case on the  $2 \times 2 \times 2$  lattice, where the degrees are 4, 3, 4, 3, 4, 3, 4, 3, sixteen line-graph edges carry sign product +1, and the worst off-diagonal block has  $\max \lambda^+ = +6.357e-01$  [h13 §1E] — exactly the  $+W_k > 0$  of step (1) = (2).

**Corollary C.2.2 (scalar form, and the normal-angle condition C3).** *Let all bodies be point masses ( $d_k = 3, M_k^{-1} = (1/m_k) I_3$ ), let each contact carry its normal row only ( $m_c = 1, B_{ck} = n_{c^T}$ ), and assume (C1). Then*

$$\begin{aligned} G_{\{cc'\}} &= \sigma_{ck} \sigma_{\{c'k\}} (1/m_k) n_{c^T} n_{\{c'\}} & (c \neq c', \text{ sharing body } k), \\ G_{\{cc\}} &= \sum_{\{k \in \{a_c, b_c\}\}} 1/m_k, \end{aligned}$$

so that under condition (3) of Theorem C.2 the off-diagonal is  $-(1/m_k) n_c \cdot n_{\{c'\}}$ , which is  $\leq 0$  if and only if  $n_c \cdot n_{\{c'\}} \geq 0$ . Hence  $G = B W B^T$  with the block incidence  $B_{\{ck\}} := \sigma_{ck} n_{c^T} \in R^{\{1 \times 3\}}$  and  $W := \text{blkdiag}((1/m_k) I_3)$ , with the static three-column block dropped. If all normals are parallel, this reduces to the scalar signed incidence  $B_{\{ck\}} = \sigma_{ck}$  and  $W = \text{diag}(1/m_k)$ , so  $G$  is a grounded weighted scalar Laplacian.

*Proof.* Lemma C.1 with  $B_{ck} = n_{c^T}$  and  $M_k^{-1} = (1/m_k) I_3$  gives  $\sigma_{ck} \sigma_{\{c'k\}} n_{c^T} (1/m_k) I_3 n_{\{c'\}} = \sigma_{ck} \sigma_{\{c'k\}} (1/m_k) n_c \cdot n_{\{c'\}}$ , a scalar; the diagonal uses  $n_{c^T} n_c = 1$ . With  $\sigma_{ck} \sigma_{\{c'k\}} = -1$  the sign of the off-diagonal is the sign of  $-n_c \cdot n_{\{c'\}}$ , so negativity is equivalent to  $n_c \cdot n_{\{c'\}} \geq 0$ , i.e. to a relative normal angle of at most  $90^\circ$ . Finally, put  $B_{\{ck\}} := \sigma_{ck} n_{c^T} \in R^{\{1 \times 3\}}$  and  $W := \text{blkdiag}((1/m_1) I_3, \dots, (1/m_{\{n_b\}}) I_3)$ , so that  $B \in R^{\{n_c \times 3n_b\}}$  and  $G = B W B^T$  is Corollary C.1.1 specialised to  $1 \times 1$  row blocks.  $B$  reduces to the scalar signed incidence  $B_{\{ck\}} = \sigma_{ck}$  with  $W = \text{diag}(1/m_k)$  exactly when all normals are parallel, which is the hypothesis under which §2.1’s scalar statement is written. ■

This is C3 and the first line of §2.1, and it is the exact statement the corpus measures: off-diagonal  $0.0$  at  $89^\circ$  and  $5.0e-1$  at  $120^\circ$  on the tilted 3-chain [08 §1]. Under patch lumping the condition disappears, and Theorem C.2 says why: the angle enters only the gauge  $D_c$  (through  $E_c$ ), never the conductance  $W_k$ , and the sign of an off-diagonal *block* is decided by  $\sigma_{ck} \sigma_{\{c'k\}}$  alone. Measured: the  $2 \times 2 \times 2$  lattice has patch normals  $90^\circ$  apart on sign-product -1 edges and those blocks are negative semidefinite anyway,  $\max \lambda^+ = -2.7986e-03$  [h13 §1D].

**Corollary C.2.3 (rotation — C4).** *Rotation never breaks the block form.* In the common gauge the contact’s kinematic block is  $Ad_k$ , which depends on the body’s position but not on the contact’s direction data, so rotation enters  $G$  only through  $W_k = Ad_k M_k^{-1} Ad_k^T$ , which is SPD by Lemma C.0; the incidence  $\sigma$  is untouched. In the scalar form of Corollary C.2.2 the same coupling appears as the off-diagonal  $(r \times n)^T I^{-1} (r' \times n')$ , a scalar of unrestricted sign — a scalar is negative semidefinite only if it is  $\leq 0$ , so a single sign flip destroys the property, whereas a  $6 \times 6$  block absorbs the same coupling inside  $-W_k \leq 0$ . Measured:  $\|G_{\{n,t1\}}\|_\infty / \|G_{\{nn\}}\|_\infty = 0.1875$  scalar [08 §1] against off-diagonal block =  $-W_{\text{shared}}$  to  $0.000e+00$  [h13 §1C].

### C.3 The friction extension (statement (iii))

**Proposition C.5 (friction rows,  $G = L \otimes I_3$ ).** *Let all bodies be point masses,  $M_k^{-1} = (1/m_k) I_3$ , and let each contact carry its three rows in its own orthonormal frame  $E_c = [n_c; t1_c; t2_c] \in 0(3)$ , i.e.  $B_{ck} = E_c$ . Then for all  $c, c'$*

$$G_{\{cc'\}} = L_{\{cc'\}} \cdot E_c E_{\{c'\}}^T, \quad L_{\{cc'\}} := \sum_k \sigma_{ck} \sigma_{\{c'k\}} / m_k,$$

with  $L$  the scalar (grounded) Laplacian of Corollary C.2.2. If in addition all contacts share one frame,  $E_c \equiv E$  (in particular if all normals are parallel and the tangent frames are chosen alike), then

$$G = L \otimes I_3,$$

so that  $G_{\{nn\}} = G_{\{t1t1\}} = G_{\{t2t2\}} = L$ , every normal-tangential and tangential-tangential cross block vanishes identically, and  $\text{spec}(G) = \text{spec}(L)$  with every multiplicity tripled. Consequently friction multiplies the row count by three without changing the operator's spectral content or its Laplacian status; what friction changes is the cone  $K_\mu$ , not  $G$ .

*Proof.* By Lemma C.1,  $G_{\{cc'\}} = \sum_k \sigma_{ck} \sigma_{\{c'k\}} E_c (1/m_k) I_3 E_{\{c'\}}^T = (\sum_k \sigma_{ck} \sigma_{\{c'k\}}/m_k) E_c E_{\{c'\}}^T$ , the scalar  $1/m_k$  commuting out of the product precisely because the inverse inertia is **isotropic** — this is the one place where  $M_k^{-1} = (1/m_k) I_3$  is used, and it is what “isotropic per-contact frames” means in the statement. The bracket is  $L_{\{cc'\}}$  by definition. With  $E_c \equiv E$ ,  $E_c E_{\{c'\}}^T = E E^T = I_3$ , hence  $G_{\{cc'\}} = L_{\{cc'\}} I_3$ , which is the block form of  $G = L \otimes I_3$ . The three stated consequences are immediate: selecting the (1,1), (2,2), (3,3) entries of each block gives  $G_{\{nn\}} = G_{\{t1t1\}} = G_{\{t2t2\}} = L$ , the off-diagonal entries within each block are  $L_{\{cc'\}} \cdot 0 = 0$ , and  $\text{spec}(A \otimes I_3) = \text{spec}(A)$  with multiplicities multiplied by 3 (if  $Ax = \lambda x$  then  $(A \otimes I_3)(x \otimes u) = \lambda (x \otimes u)$  for each of three independent  $u \in \mathbb{R}^3$ , and these  $3n_c$  vectors form a basis). ■

*Measured counterpart:*  $\|G_{nn} - G_{t1t1}\|_\infty = \|G_{nn} - G_{t2t2}\|_\infty = 0$  and all cross blocks exactly 0 [08 §1]. *General frames:* when the  $E_c$  differ,  $R_{\{cc'\}} := E_c E_{\{c'\}}^T \in O(3)$  and  $G = (L_{\{cc'\}} R_{\{cc'\}})_{\{cc'\}}$  is a **connection (gain-graph) Laplacian** with orthogonal connection: the diagonal is unchanged and  $\|G_{\{cc'\}}\|_2 = \|L_{\{cc'\}}\|_2$  because  $R_{\{cc'\}}$  is an isometry. The off-diagonal block is not symmetric, so the sign test is applied to  $\text{sym}(-(1/m_k) R_{\{cc'\}})$ , the convention used in Theorem C.2 step (1)  $\Rightarrow$  (2) and in the Frobenius share of Figure 2. With that convention the block is negative semidefinite if and only if  $\text{sym}(R_{\{cc'\}}) \geq 0$ , i.e.  $R_{\{cc'\}}$  is a rotation of angle at most  $90^\circ$ ; the normal-normal entry of that condition is  $n_c \cdot n_{\{c'\}} \geq 0$ , which is Corollary C.2.2's angle condition. Patch lumping replaces  $R_{\{cc'\}}$  by the gauge  $D_c$ , which is applied by congruence; by Corollary C.1.1 that congruence preserves  $-W_k \leq 0$  only in the common gauge, and the verdict is read there.

#### C.4 The spectral identity (statement (iv), Corollary 1 of §2.4)

**Theorem C.3 (spectrum).** Let  $M > 0$  and let  $J$  be **any** contact Jacobian — no hypothesis (C1)–(C6), no gauge assumption, any mixture of row types. Put  $K := J M^{-1/2}$ , with  $M^{1/2}$  the symmetric positive definite square root of  $M$ . Then

$$G = K K^T, \quad L_K := K^T K = M^{-1/2} J^T J M^{-1/2}, \quad L_v := J^T J,$$

and

1.  $\text{spec}(G) \setminus \{0\} = \text{spec}(L_K) \setminus \{0\}$ , with equal multiplicities;
2.  $\mu \neq 0$  lies in  $\text{spec}(L_K)$  if and only if  $\mu$  is a generalized eigenvalue of the pencil  $L_v x = \mu M x$  with  $L_v := J^T J$ ;
3. in the common gauge  $(J^T J)_{\{kl\}} = \sum_c \sigma_{ck} \sigma_{cl} \text{Ad}_k^T \text{Ad}_l$ , i.e.  $J^T J$  is the block **vertex Laplacian** of the contact graph (bodies as vertices);
4.  $\text{rank } G = \text{rank } J$ , so  $G$  has exactly  $(\sum_c m_c) - \text{rank } J$  zero eigenvalues.

*Proof.*  $M^{1/2}$  exists, is SPD and invertible, by the spectral theorem applied to  $M > 0$ . Then  $K K^T = J M^{-1/2} M^{-1/2} J^T = J M^{-1} J^T = G$ , and  $L_K = K^T K = M^{-1/2} J^T J M^{-1/2}$  by definition.

1. Let  $\lambda \neq 0$  and let  $V_\lambda = \ker(KK^\top - \lambda I)$ ,  $U_\lambda = \ker(K^\top K - \lambda I)$ . The map  $\Phi : x \mapsto K^\top x$  sends  $V_\lambda$  into  $U_\lambda$ , since  $K^\top K(K^\top x) = K^\top(KK^\top x) = \lambda K^\top x$ . It is injective on  $V_\lambda$ : if  $x \in V_\lambda$  and  $K^\top x = 0$  then  $\lambda x = K K^\top x = 0$ , and  $\lambda \neq 0$  forces  $x = 0$ . Hence  $\dim V_\lambda \leq \dim U_\lambda$ . The symmetric map  $\Psi : y \mapsto K y$  sends  $U_\lambda$  into  $V_\lambda$  and is injective there by the same argument, giving  $\dim U_\lambda \leq \dim V_\lambda$ . So  $\dim V_\lambda = \dim U_\lambda$  for every  $\lambda \neq 0$ ; in particular one is nonzero exactly when the other is. Both matrices are real symmetric, so geometric and algebraic multiplicities coincide, and the nonzero spectra agree as multisets.
2.  $K^\top K x = \mu x$  reads  $M^{-1/2} J^\top J M^{-1/2} x = \mu x$ . Put  $y := M^{-1/2} x$ , a bijection of  $R^{n_v}$  since  $M^{-1/2}$  is invertible. Left-multiplying by  $M^{1/2}$  gives  $J^\top J y = \mu M^{1/2} M^{1/2} y = \mu M y$ . The steps reverse, so the correspondence  $x \leftrightarrow y$  is a bijection between the eigenspaces of  $K^\top K$  and the generalized eigenspaces of  $(J^\top J, M)$  for each  $\mu$ , zero included.
3. By (N3),  $J^\top J = \sum_c J_{c^\top} J_c$ , whose  $(k, l)$  block is  $\sum_c \sigma_{ck} \sigma_{cl} B_{ck}^\top B_{cl} = \sum_c \sigma_{ck} \sigma_{cl} \text{Ad}_{k^\top} \text{Ad}_l$  in the common gauge — the vertex-space counterpart of Corollary C.1.1, with the roles of contacts and bodies exchanged.
4.  $\text{rank } G = \text{rank } KK^\top = \text{rank } K = \text{rank } (J M^{-1/2}) = \text{rank } J$ , using  $\text{rank } AA^\top = \text{rank } A$  for real  $A$  and the invertibility of  $M^{-1/2}$ . ■

Three consequences worth stating separately, because §2.4 and §6.3 depend on exactly this separation:

- **The identity is independent of C1-C6.** Only  $M > 0$  is used. In particular it survives body degree  $\geq 3$ : the condition of Theorem C.2 governs whether the *assembly* is a Laplacian (sign structure), not whether the *spectrum* is readable. Measured: the  $2 \times 2 \times 2$  lattice, where the sign structure fails, is estimated as accurately as the chains (rel err  $1.2e-15$ ) [h15 §1A, §1D]; and the dense signed assembly equals  $G$  to  $2.4e-17$  there [h15 §1E].
- **It is gauge-local.**  $G_{\text{loc}} = D G_{\text{common}} D^\top$  is a congruence, so  $\text{spec}(G_{\text{loc}}) \neq \text{spec}(G_{\text{common}})$  in general; Theorem C.3 must be applied with  $L_v = J^\top J$  formed from the *same*  $J$  whose  $G$  is wanted. Measured:  $\lambda_{\min^+}$  differs by up to  $26\times$  between gauges [h15 §1F].
- **It is what makes the estimator matrix-free.**  $K$  and  $K^\top$  are applied as two sparse products without forming  $G$ , and the pencil side is  $n_v \times n_v$  rather than  $(\sum_c m_c) \times (\sum_c m_c)$ . Measured:  $\leq 204$  matvecs at relative error  $1.2e-15 \dots 1.8e-9$  [h15 §1A], and 2.5 % of the solve's matvec budget at  $n_c = 4096$  [h1 §H1b.1].

## C.5 Patch lumping (statement (v), Corollary 2 of §2.5)

Throughout this section, one face-face patch: reference point  $p_0$  (the patch centroid), right-handed orthonormal frame  $(t_1, t_2, n)$  with  $n = t_1 \times t_2$ , half-extents  $e_1 > 0$  along  $t_1$  and  $e_2 > 0$  along  $t_2$ , and the four coplanar corner points

$$p^{\{st\}} = p_0 + s e_1 t_1 + t e_2 t_2, \quad s, t \in \{-1, +1\}, \quad r^{\{st\}} := p^{\{st\}} - p_0.$$

The patch belongs to a body with centre  $c_k$ ; write  $r := p_0 - c_k$ . Gauge scalars  $S = \text{diag}(1, e_2, e_1, 1, 1, \ell)$ ,  $\ell = \sqrt{(e_1^2 + e_2^2)/3}$ , and the wrench ordering  $w = (F_n, M_{t_1}, M_{t_2}, F_{t_1}, F_{t_2}, M_n)$  of §2.1. Two elementary cross products are used repeatedly and follow from  $n = t_1 \times t_2$  and the Jacobi/triple-product identity  $a \times (b \times c) = b(a \cdot c) - c(a \cdot b)$ :

$$\begin{aligned} t_1 \times n &= t_1 \times (t_1 \times t_2) = t_1(t_1 \cdot t_2) - t_2(t_1 \cdot t_1) = -t_2, \\ t_2 \times n &= t_2 \times (t_1 \times t_2) = t_1(t_2 \cdot t_2) - t_2(t_2 \cdot t_1) = +t_1. \end{aligned}$$

**Lemma C.6 (point Jacobian: rank 6 of 12).** *The 12 point-contact rows of one 4-corner patch on a single rigid body span exactly a 6-dimensional space; hence  $\text{rank } J_{\text{points}} = 6$ ,  $\text{rank } G_{\text{points}} = 6$ , and  $G_{\text{points}} \in R^{12 \times 12}$  is singular.*

*Proof.* Each row is  $(d^\top, (\rho_j \times d)^\top)$  with  $d \in \{n, t1, t2\}$  and  $\rho_j = p^j - c_k = r + r^j$  (h13\_patch.py:J\_points). The row space is a subspace of  $R^6$ , so  $\text{rank} \leq 6$ ; only the reverse inclusion needs work. Differences of two rows with the same  $d$  at two corners give  $(0, ((\rho_j - \rho_l) \times d)^\top) = (0, ((r^j - r^l) \times d)^\top)$ . Taking the two corner pairs that differ in  $s$  and in  $t$  respectively yields the available differences  $\Delta_1 = 2 \text{ e1 } t1$  and  $\Delta_2 = 2 \text{ e2 } t2$ , and then

$$\begin{aligned}\Delta_1 \times n &= 2 \text{ e1 } (t1 \times n) = -2 \text{ e1 } t2, \\ \Delta_1 \times t2 &= 2 \text{ e1 } (t1 \times t2) = +2 \text{ e1 } n, \\ \Delta_2 \times n &= 2 \text{ e2 } (t2 \times n) = +2 \text{ e2 } t1,\end{aligned}$$

so  $(0, t1), (0, t2), (0, n)$  all lie in the row space (using  $\text{e1}, \text{e2} > 0$ ), i.e.  $\{0\} \times R^3 \subseteq \text{rowspace}$ . Reducing the original rows  $(d, \rho_j \times d)$  modulo  $\{0\} \times R^3$  leaves  $(d, 0)$  for  $d = n, t1, t2$ , so  $R^3 \times \{0\} \subseteq \text{rowspace}$  as well. Hence the row space is all of  $R^6$  and  $\text{rank } J\_points = 6$ . Finally  $\text{rank } G\_points = \text{rank } (J\_points M^{-1} J\_points^\top) = \text{rank } J\_points = 6 < 12$  by Theorem C.3(4), so  $G\_points$  is singular and its condition number over the full matrix is infinite. ■

(Measured:  $\text{rank } J\_points = 6$  of 12 rows,  $\text{cond } G\_points = \infty$ , at both  $\mu = 0$  and  $\mu = 0.5$  [h13 §3A].)

**Corollary C.6.1 (the point-patch solution is not unique; the lumped margin is).** *On a 4-corner point patch the bilateral solution is not unique — rank  $J\_points = 6$  of 12 by Lemma C.6, so  $G\_points$  is singular — and the regularised solve  $\lambda = (G + \eta I)^{-1}(-J \text{ v\_free})$  returns the minimum-norm element of the 6-dimensional self-stress space of Lemma C.8; the certificate margin computed on that element therefore depends on the solver’s selection. After lumping, the centre of pressure is unique (rank  $C = 6$ ,  $\ker C =$  the self-stress space, Lemma C.8) and the margin coincides with the polar radius of the wrench-cone section. The agreement with a measured limit is a function of the load rate, not a step-load restriction: the error of the cone section against the measured bracket falls monotonically with the ramp time  $\tau$ , 24.0 % at  $\tau = 1 \text{ s}$  and 2.7 % at  $\tau = 20 \text{ s}$  [U134 §“Ramp- och stegresultat”], and the quantity the bracket locates is the onset of sliding, not tipping [A74 §4].*

*Measured, lumped side.* Writing the patch cone of Theorem C.9 in physical units places the centre of pressure inside the rectangle, and the limit against a lateral body-force load is  $L(\varphi) = \min(\mu_i g, \mu_g g, (g/h_i) r_{\text{rect},i}(\varphi), (g/h_g) r_{\text{rect},g}(\varphi))$  with  $r_{\text{rect}}$  the rectangle’s polar radius and  $h$  the centre-of-mass height over the patch — no fitted parameter. The load is a constant lateral acceleration applied as a body force on every body from the first step,  $dt = 0.005 \text{ s}$ , 120 steps, horizon 0.6 s, fall at 5 mm relative or ground drift or  $5^\circ$  tilt [U126 §1]. Against a forward simulation bracketed in 24 directions on three scenes (box alone, two-box stack, pallet with an overhanging crate) the mean relative error is 0.66 %, 0.66 % and 1.13 %, the max is 2.3 %, and the fall mechanism is the predicted one in 24 of 24 directions in all three scenes [U126 §9]. The centre-of-pressure law  $p_{\text{CoP}} = \text{centre-of-mass projection} + (h L/g) \varphi$  is measured with max absolute error  $6.06\text{e-}10 \text{ m}$  (box, ground patch),  $6.63\text{e-}08 \text{ m}$  (stack, interface patch) and  $8.01\text{e-}04 \text{ m}$  (pallet, ground patch) over 24 directions [U126 §5]. Where friction does not bind, the boundary is the rectangle’s polar radius: at  $\mu = 1.50$  the same model gives 2.0 %, 0.8 % and 1.0 % mean relative error, the pieces are the four sides, and the single carrying corners occur at  $\varphi \pm 45^\circ, \pm 135^\circ$ , largest distance between predicted corner direction and measured switch  $7.5^\circ$  (square ground patch) and  $9.4^\circ$  (rectangular ground patch), both below the  $15^\circ$  direction resolution [U126 §9, §5]. The seed rectangle without the friction branch gives 54.7 %, 86.5 % and 70.5 % — 0 of 3 scenes under 10 % [U126 §9].

*Measured, point side.* The certificate’s polar curve  $m_{\text{cert}}(S_0)$ , built on the nominal active set with  $\eta = 1\text{e-}6 \cdot \text{mean}(\text{diag } G)$  and U45’s combined margin, lies outside the measured bracket in **144** of **144** direction-scene pairs, mean relative error 0.198 ... 2.979, max 6.053; in the pallet it is negative in all 24 directions already at the 0.20 g reference load while the forward simulation holds to 2.21–3.97 m/s<sup>2</sup> [U126 §6].  $L_{\text{cert}}$  moves at most  $6.64\text{e-}04$  relative when  $\eta$  is varied by a

factor 10 in both directions, so the deviation is a property of the selection, not of the regularisation [U126 §6].

*Measured, enumerated side.* Enumerating every subset  $S$  of the contact candidates statically — pruning those without a support path to ground, then testing  $J_n^\top \lambda_n = f_0$  by LP — and taking the polar radius of the union by SOCP ( $\max L \geq 0$  under  $J^\top \lambda = f_0 + L F d$  and the Coulomb cone) reproduces the same step-load bracket over **72 directions per scene** with mean relative error 0.658227855 % (box), 0.658227855 % (stack) and 1.096361489 % (pallet), max 2.248681837 %, without a fitted parameter [U131 §“Mätning och acceptans”]. Candidate contacts 4 / 8 / 8; weight-bearing subsets 7 / 49 / 9; subsets that also pass the regularised bilateral cone test 7 / 25 / 9; observed active sets 1 / 1 / 12 [U131, summary lines 4–8]. The SOCP radius agrees with the analytic cut to 2.233e–09, 1.369e–09 and 1.908e–09 absolute, and the LP bound width is at most 4.672921e–03 [U131 §“Facetter och styckegraf”]. The certificate and the radius separate on the same  $S_0$ : mean relative error 37.989732 %, 16.558203 % and 72.240712 % for  $L_{\text{cert}}(S_0)$  against 0.658228 %, 0.658228 % and 32.711811 % for the radius over the dynamically observed sets, the last figure carrying U131’s rule that a set which does not bear the nominal weight is assigned radius 0 rather than a positive margin [U131 §“Certifikat och observerade S”].

*What the enumerated form is not.* With circular Coulomb friction the section is not in general polyhedral: the box and the stack give circles of radius  $\mu_{\min} g$  (4.905000000 and 2.452500000 m/s<sup>2</sup>) and the pallet’s boundary is one line segment ( $a_x \leq 2.212981002$  m/s<sup>2</sup>) plus a circular arc, so a finite graph of line segments covers none of the three boundaries; the LP approximations are cycles of 72, 72 and 51 facets [U131 §“Facetter och styckegraf”]. The proposed identity for  $m_{\text{cert}}$  does not follow from the construction: the regularised bilateral solver selects one force distribution while the wrench cone admits every cone-feasible one [U131 §“Kontroller och begränsningar”]. Box and stack carry one observed active set each, so the enumeration is exercised on the pallet alone (12 dynamic nodes, 38 edges, 3 of them pruned as not weight-bearing) [U131 §“Certifikat och observerade S”]. The three scenes are single bodies and two-body stacks on horizontal rectangular patches under a body-force load; new body pairs and other geometries are not validated by that acceptance [U131 §“Kontroller och begränsningar”]. PGS keeps U126’s budget of 60 sweeps at tolerance 1e–8 and the max NCP residuals are 2.29018013e–05, 1.18817623e–05 and 0.000159298178, so the tolerance is not reached at every step of the inherited bracket [U131 §“Kontroller och begränsningar”]. Bit identity: 220 artifact pairs equal byte for byte, physics payload sha256 ac70d5cce982638460606edb97c098cbddb99cf1083d62fd479664298d5f3897 in both rounds, 6 artifacts recomputed byte-identical in a new process [U131 §“Bitidentitet, tid och misslyckanden”]. U126’s bracket bits, profile bits and certificate reproduce over two processes with full digests, and its U43b runs exceed the lane’s 90 s limit, which U126 records as a deviation [U126 §8].

*Audit, and the load regime the result is bounded to.* A73 re-ran U131’s statics from copies of its code in two new processes: 7 / 49 / 9 weight-bearing subsets and 72 / 72 / 51 LP facets in both, sha256 12ea2e0b51a41b591f9750223a4c09d9d498c9fb286966409dbe2f554d714589 in both, and the published 0.658227855 % / 0.658227855 % / 1.096361489 % reproduced [A73 §1]. Changing the load rule breaks the agreement. Under a linear ramp  $L_s = L (s+1)/150$  with  $dt = 0.004$  s, 150 steps, horizon 0.6 s and eight bisections over 24 directions, the static radius 4.905000 / 2.452500 / 2.212981–3.924000 m/s<sup>2</sup> meets a ramp bracket of 6.3671875 / 3.41796875 / 3.44921875–5.1953125 m/s<sup>2</sup>: mean relative error **22.9644 %**, **28.2469 %** and **27.8781 %**, max 36.9146 % [A73 §2]. So the reading of this corollary is a load-rate law on the slip branch, not a step-load restriction. The error of the enumerated cone section against the measured bracket falls monotonically with the ramp time  $\tau$ : 442.16 %, 94.23 %, 24.03 %, 7.49 % and 2.73 % at  $\tau = 0.05$ , 0.2, 1, 5 and 20 s over 72 direction-scene pairs, and 2.414628 % (box), 3.051733 % (stack) and 2.727546 % (pallet) at  $\tau = 20$  s [U134 §“Ramp- och stegresultat”]. The audit continues the ramp past U134’s last point: at  $\tau = 50$  s and  $\tau = 100$  s the error in direction 0 of the box scene is 1.458970 % and 0.821865 % [A74 §2]. A74 reproduced U134 byte for byte in new processes,

sha256 6fb4b0de95aba90bee5ca09da50c7f294c4047079ae299fbc982b1558a669346 at  $\tau = 1$  s and 20975aa50e022baa6b090f1f115c41da491561c8ba39405981e7fd638c2d0ce9 at  $\tau = 20$  s [A74 §1]. The constant step load gives 0.503313 %, 1.140418 % and 0.724155 % on the same three scenes [U134 §“Ramp- och stegresultat”], so the slowest ramp U134 measured is still a factor 3–5 above the step result, and the 50 s and 100 s points close that gap monotonically. A73’s 22.96 / 28.25 / 27.88 % is a ramp of  $\tau = 0.6$  s reported as  $|r - L|/L$  while U134 reports  $|L - r|/r$ , so the two percentages are not the same measure; A74 re-ran  $\tau = 0.6$  s under U134’s rule and obtained 6.3515625 / 3.41015625 / 3.44140625 m/s<sup>2</sup> against A73’s 6.3671875 / 3.41796875 / 3.44921875 m/s<sup>2</sup> in direction 0 [A74 §3].

*Which limit the bracket measures.* The quantity located is the onset of sliding, not tipping. At  $\tau = 20$  s in direction 0 of the box scene the slip-as-fall bracket is [5.015625, 5.03125] m/s<sup>2</sup>, midpoint 5.0234375 m/s<sup>2</sup>, while tip-as-fall returns no bracket through the largest load sampled, 32 m/s<sup>2</sup>: at that load sliding starts after 3.330 s, the final drift is 1 296.168246 m and the maximum tilt is 1.215701e−06 rad [A74 §4]. U134’s “432 of 432” and “72 of 72” therefore record that slip started before tipping, not that a tipping limit was measured; the separate tipping bracket was run for the box scene, direction 0, only, and no tipping limit is claimed for the stack or the pallet [A74 §4]. The polar radius of the wrench-cone section on these three flat-support scenes is  $\min(\mu g, g \cdot a/h)$ , with  $a$  the support half-extent in the load direction and  $h$  the centre-of-mass height over the patch; an independent construction of that expression from the scene bodies’ dimensions differs from U131’s SOCP radius by at most 2.232757979e−09 m/s<sup>2</sup> over 72 directions [A74 §5]. Doubling the mass leaves the  $\tau = 20$  s bracket [5.015625, 5.03125] m/s<sup>2</sup> unchanged, and halving  $\mu$  gives [2.5234375, 2.53125] m/s<sup>2</sup>, load ratio 0.503110420 [A74 §6]. The statement this corollary carries is therefore: the cone section’s polar radius is  $\min(\mu g, g \cdot a/h)$ ; the slip branch is the quasi-static limit, reached to 1.46 % at  $\tau = 50$  s and 0.82 % at  $\tau = 100$  s in the isotropic box scene; the tip branch is verified only at  $\mu = 1.50$ , where friction does not bind and the model gives 2.0 %, 0.8 % and 1.0 % mean relative error [U126 §9]. The residual 2.7 % at  $\tau = 20$  s is reported at one fall definition; the threshold sweep was run at  $\tau = 5$  s only, where the definition moves the error over 4.644–10.378 % [U134 §“Falldefinition vid  $\tau=5$  s”]. Under U134’s own 3 % criterion the combined figure passes and the stack does not, at 3.051733 % [U134 §Verdikt]. Bit identity: sha256 over sorted path plus contents of all 432 main files and 72 threshold files is 6017f06f4b49b8ccdd34f2d1e604dea8b05f65c3d6bb67dcb466da83896f08a3, and the maximum PGS final residual is 1.5929817836033734e−04 against a tolerance of 1e−08, so the tolerance is not reached at every step [U134 §“Bitidentitet och kontroller”].

Four further checks fall and one passes. At  $\mu = 0.1$  the frozen-geometry radius is 0.981000 m/s<sup>2</sup> against a step bracket of 1.009765625 m/s<sup>2</sup> in all 24 directions — 2.8488 %, outside the 0.7–1.1 % band [A73 §4]. Doubling both the nominal weight wrench  $f_0$  and the acceleration map  $F$  moves the acceleration radius by at most 2.473431e−09 m/s<sup>2</sup>, 5.043e−10 relative [A73 §5]. A circular-bottom body has zero static restoring moment about every contact point, so the ideal radius is 0 while a frozen four-point model with  $\mu = 0.1$  gives 0.981000 m/s<sup>2</sup>; this is an analytic counterexample and no discrete rounded-SDF simulation was run [A73 §4]. The gripper transfer fails as a definition, not numerically: U45’s gripper gives 8 contacts and 209 weight-bearing subsets of 256, U70’s STEP 00 gives 209 of 256 and STEP 18 gives 0 of 256, but the cone carries no upper bound on normal force, so a force-closed grasp has unbounded radius while a bracket against the prescribed 25 N clamp force is finite [A73 §3]. The enumeration does not scale:  $n_c = 48$  needs  $2^{48} = 281474976710656$  subsets and  $72 \cdot 2^{48} = 20266198323167232$  LP solves for 72 directions, and when the full set is admissible the enumeration is redundant for the outer boundary of the union, since  $K(S) \subseteq K(C)$  [A73 §6; U131 §“Representation och definitioner”].

*Relation to §4.* §4 reads the certificate as the stability margin on the stated patch-lumped K8 scenes, where its zero crossings coincide with the ZMP and Coulomb margins; the monotone one-dimensional sweep’s  $r = 1$  is non-diagnostic, and 40 randomized towers reach pointwise discrepancy 0.97 [U22 §2.1, §5; A81]. This corollary bounds that reading: on point patches the

margin the certificate reports is the margin of a minimum-norm selection and misses the measured bracket in 144 of 144 direction-scene pairs [U126 §6].

**Lemma C.7 (patch Jacobian: full row rank 6).** *The  $6 \times 6$  block of  $J_{\text{patch}}$  at a dynamic body is invertible. Consequently  $J_{\text{patch}}$  has full row rank 6 per patch, and if the scene's  $J_{\text{patch}}$  has full row rank then  $G_{\text{patch}} = J_{\text{patch}} M^{-1} J_{\text{patch}}^T$  is symmetric positive definite; in particular the  $6 \times 6$  diagonal block of the lumped  $G$  at any patch is SPD, so the lumped  $G$  restricted to the patch is nonsingular.*

*Proof.* From `h13_patch.py:_patch_rows_local`, the block has three **force rows**  $(d, r \times d)$  for  $d = n, t1, t2$  and three **moment rows**  $(0, \sigma d)$  for  $(d, \sigma) = (t1, e2), (t2, e1), (n, \ell)$ . Order the rows as (force rows; moment rows) and split the columns into (linear; angular). The block becomes

$$\begin{bmatrix} E & X \\ 0 & \Sigma P E \end{bmatrix}$$

with  $E = [n; t1; t2] \in 0(3)$  (the frame, as rows),  $X$  the  $3 \times 3$  matrix with rows  $(r \times d)^T$ ,  $\Sigma = \text{diag}(e2, e1, \ell)$  and  $P$  the permutation sending  $(n, t1, t2)$  to  $(t1, t2, n)$ . This is block upper triangular, so its determinant is  $\det(E) \cdot \det(\Sigma P E) = \det(E)^2 \det(\Sigma) \det(P) = \pm e1 e2 \ell \neq 0$  because  $e1, e2 > 0$  (hence  $\ell > 0$ ) and  $|\det E| = 1$ . Therefore the block is invertible and the six rows are independent — already within the columns of a single body, so a patch between two dynamic bodies has full row rank a fortiori. Row permutation and the gauge  $S$  change nothing (both invertible).

For the second claim, let  $J_{\text{patch}}$  have full row rank. For  $x \neq 0$ ,  $x^T G_{\text{patch}} x = (J_{\text{patch}}^T x)^T M^{-1} (J_{\text{patch}}^T x) > 0$ , since  $J_{\text{patch}}^T$  is injective (full row rank of  $J_{\text{patch}}$ ) and  $M^{-1} > 0$ . Hence  $G_{\text{patch}} > 0$ . The diagonal block at patch  $c$  is  $J_c M^{-1} J_c^T$ , which is SPD by the same argument applied to the single row block. ■

(Measured: rank  $J_{\text{patch}} = 6$  of 6 and cond  $G_{\text{patch}} = 8.550e+00$  on the pushed cube [h13 §3A]; 18/18 and 48/48 with finite condition number on the column and the K8 tower [h13 §1H]. The caveat of §2.5 is consistent with the lemma: on the  $2 \times 2 \times 2$  lattice  $J_{\text{patch}}$  is 96 rows of rank 48, because full row rank per patch does not imply full row rank of the stacked  $J_{\text{patch}}$  when loops make the scene hyperstatic at the wrench level.)

**Lemma C.8 (the lumping map and the wrench identity).** *Define  $C$  on the corner impulses of a patch by the gauged resultant at  $p_0$ :*

$$\begin{aligned} f &:= \sum_j f_j, & m &:= \sum_j r^j \times f_j, & f_j &\in \mathbb{R}^3 \text{ the impulse at corner } j, \\ (C\lambda)_0 &= n \cdot f, & (C\lambda)_3 &= t1 \cdot f, & (C\lambda)_4 &= t2 \cdot f, \\ (C\lambda)_1 &= (t1 \cdot m)/e2, & (C\lambda)_2 &= (t2 \cdot m)/e1, & (C\lambda)_5 &= (n \cdot m)/\ell, \end{aligned}$$

(this is `h13_patch.py:lumping_matrix` column by column). Then

$$J_{\text{points}}^T \lambda = J_{\text{patch}}^T (C \lambda) \quad \text{for every } \lambda,$$

rank  $C = 6$ , and  $\ker C = \ker J_{\text{points}}^T$  is exactly the 6-dimensional self-stress space of the patch.

*Proof.* Fix a body with centre  $c_k$ . A point impulse  $f_j$  applied at  $p^j = p_0 + r^j$  contributes the generalized impulse  $(f_j, (p^j - c_k) \times f_j) = (f_j, r \times f_j + r^j \times f_j)$  where  $r = p_0 - c_k$ ; summing over corners, the point model contributes  $(f, r \times f + m)$ . Transposing `h13_patch.py:_patch_rows_local`, the patch model maps a gauged wrench  $w$  to  $(f(w), r \times f(w) + m(w))$  with  $f(w) = w_0 n + w_3 t1 + w_4 t2$  and  $m(w) = w_1 e2 t1 + w_2 e1 t2 + w_5 \ell n$  — the force rows contribute  $(d, r \times d)$  per unit of the corresponding component and the moment rows contribute  $(0, \sigma d)$ , which is exactly the statement that the patch row block realises the pair (resultant force, resultant moment about  $p_0$ ). By construction  $f(C\lambda) = f$  and  $m(C\lambda) = m$ , since the

six defining equations of  $C$  are the components of  $f$  in the frame and of  $m$  in the frame divided by the gauge scalars, and  $(n, t_1, t_2)$  is an orthonormal basis. Hence both models produce the same  $(f, r \times f + m)$  at each incident body, i.e.  $J_{\text{points}}^T \lambda = J_{\text{patch}}^T (C\lambda)$ . (Measured:  $1.3e-16 \dots 2.0e-16$  relative [h13 §0].)

For the rank:  $J_{\text{patch}}^T$  is injective by Lemma C.7, so  $\text{rank } C = \text{rank}(J_{\text{patch}}^T C) = \text{rank}(J_{\text{points}}^T) = 6$  by Lemma C.6, and  $\dim \ker C = 12 - 6 = 6$ . Injectivity of  $J_{\text{patch}}^T$  also gives  $C\lambda = 0 \iff J_{\text{patch}}^T C\lambda = 0 \iff J_{\text{points}}^T \lambda = 0$ , so  $\ker C$  is precisely the space of corner impulses that exert no generalized impulse on the bodies — the self-stress of the coplanar patch. ■

(Measured:  $C$  is  $6 \times 12$  of rank 6 with  $\dim \ker C = 6$  [h13 §3A]; the solution set of the point problem has measured width  $0.294300 \text{ N}\cdot\text{s}$  in  $\lambda_n$  at corner 0 while  $C\lambda$  is constant to  $2.2e-12$  over 50 warm starts [h13 §3B].)

**Theorem C.9 (the patch cone is exactly the image of the 4-corner cone).** *Restrict to frictionless corner impulses,  $f_j = \lambda_j n$  with  $\lambda_j \geq 0$ , and index the corners by  $(s, t) \in \{-1, +1\}^2$ . Then  $C$  restricted to these inputs has image in the  $(F_n, M_{\{t1\}}, M_{\{t2\}})$  coordinates and is given by the linear map*

$$A : \mathbb{R}^4 \rightarrow \mathbb{R}^3, \quad A\lambda = \sum_{\{s,t\}} \lambda_{\{st\}} \cdot (1, t, -s)^T,$$

i.e.  $A = [[1, 1, 1, 1], [t_j], [-s_j]]$ . The map  $A$  is **onto**  $\mathbb{R}^3$ , and

$$A(\mathbb{R}^4_+) = K_{\text{nm}} := \{(F, M_1, M_2) : F \geq 0, |M_1| \leq F, |M_2| \leq F\},$$

which in physical (ungauged) units is  $F_n \geq 0$  together with the centre of pressure lying in the rectangle  $[-e_1, e_1] \times [-e_2, e_2]$  around  $p_0$ . The remaining components vanish identically:  $F_{\{t1\}} = F_{\{t2\}} = M_n = 0$ .

**Proof. The map.** With  $f_j = \lambda_{\{st\}} n$ ,  $f = (\sum \lambda_{\{st\}}) n$ , so  $n \cdot f = \sum \lambda_{\{st\}} =: F$  and  $t_1 \cdot f = t_2 \cdot f = 0$ , giving the last claim for  $F_{\{t1\}}, F_{\{t2\}}$ . For the moment,  $r^{\{st\}} \times n = (s e_1 t_1 + t e_2 t_2) \times n = s e_1 (t_1 \times n) + t e_2 (t_2 \times n) = -s e_1 t_2 + t e_2 t_1$ , hence  $m = \sum_{\{st\}} \lambda_{\{st\}} (-s e_1 t_2 + t e_2 t_1)$ , and

$$\begin{aligned} (C\lambda)_1 &= (t_1 \cdot m)/e_2 = \sum_{\{st\}} \lambda_{\{st\}} (t e_2)/e_2 = \sum_{\{st\}} t \lambda_{\{st\}} =: M_1, \\ (C\lambda)_2 &= (t_2 \cdot m)/e_1 = \sum_{\{st\}} \lambda_{\{st\}} (-s e_1)/e_1 = -\sum_{\{st\}} s \lambda_{\{st\}} =: M_2, \\ (C\lambda)_5 &= (n \cdot m)/\ell = 0 \end{aligned} \quad \text{since } r^{\{st\}} \times n \perp n \text{ for every corner.}$$

So  $C\lambda = (F, M_1, M_2, 0, 0, 0)$  with  $(F, M_1, M_2) = A\lambda$  as stated. Note that as  $(s, t)$  ranges over  $\{-1, +1\}^2$ , the pair  $(t, -s)$  ranges over all four sign patterns in  $\{-1, +1\}^2$ , so the four columns of  $A$  are exactly the four vectors  $(1, \alpha, \beta)$ ,  $\alpha, \beta \in \{-1, +1\}$ .

**Onto.** The three rows of  $A$  are  $(1, 1, 1, 1)$ ,  $(t_j)_j$  and  $(-s_j)_j$ . Over the four corners,  $\sum_j t_j = 0$ ,  $\sum_j s_j = 0$  and  $\sum_j s_j t_j = 0$ , so the three rows are pairwise orthogonal and each has norm  $2 \neq 0$ ; they are therefore linearly independent,  $\text{rank } A = 3$ , and  $A$  is surjective onto  $\mathbb{R}^3$ .

$A(\mathbb{R}^4_+) \subseteq K_{\text{nm}}$ . Let  $\lambda \geq 0$  and  $(F, M_1, M_2) = A\lambda$ . Then  $F = \sum \lambda_j \geq 0$  and, by the triangle inequality with  $|\alpha_j| = 1$ ,  $|M_1| = |\sum_j \alpha_j \lambda_j| \leq \sum_j \lambda_j = F$ , and likewise  $|M_2| \leq F$ .

$K_{\text{nm}} \subseteq A(\mathbb{R}^4_+)$ . Let  $(F, M_1, M_2) \in K_{\text{nm}}$ . If  $F = 0$  then  $|M_1| \leq 0$  and  $|M_2| \leq 0$  force  $M = 0$ , and  $\lambda = 0$  works. If  $F > 0$ , set, for  $\alpha, \beta \in \{-1, +1\}$ ,

$$\lambda_{\{\alpha\beta\}} := (F + \alpha M_1)(F + \beta M_2) / (4F) \geq 0,$$

each factor being nonnegative because  $|M_1| \leq F$  and  $|M_2| \leq F$ . Expanding  $(F + \alpha M_1)(F + \beta M_2) = F^2 + \alpha F M_1 + \beta F M_2 + \alpha \beta M_1 M_2$  and using  $\sum_{\{\alpha,\beta\}} 1 = 4$ ,  $\sum_{\{\alpha,\beta\}} \alpha = \sum_{\{\alpha,\beta\}} \beta = \sum_{\{\alpha,\beta\}} \alpha \beta = 0$ ,  $\sum_{\{\alpha,\beta\}} \alpha^2 = \sum_{\{\alpha,\beta\}} \beta^2 = 4$ :

$$\begin{aligned}\sum_{\{\alpha,\beta\}} \lambda_{\{\alpha\beta\}} &= (4F^2 + 0 + 0 + 0)/(4F) = F, \\ \sum_{\{\alpha,\beta\}} \alpha \lambda_{\{\alpha\beta\}} &= (0 + 4F M_1 + 0 + 0)/(4F) = M_1, \\ \sum_{\{\alpha,\beta\}} \beta \lambda_{\{\alpha\beta\}} &= (0 + 0 + 4F M_2 + 0)/(4F) = M_2,\end{aligned}$$

so  $\lambda = (F, M_1, M_2)$  with  $\lambda \geq 0$ . Hence the image of the positive orthant is exactly  $K_{nm}$ ; the map is onto  $R^3$  and onto  $K_{nm}$  from  $R^4_+$ , and the inclusion is an equality, not a relaxation.

**Centre of pressure.** With  $F > 0$  the centre of pressure is  $p_{CoP} = p_0 + (\sum_j \lambda_j r^j)/F$ , whose frame coordinates are

$$\begin{aligned}(p_{CoP} - p_0) \cdot t_1 &= e_1 (\sum_j s_j \lambda_j)/F = -e_1 M_2 / F, \\ (p_{CoP} - p_0) \cdot t_2 &= e_2 (\sum_j t_j \lambda_j)/F = +e_2 M_1 / F,\end{aligned}$$

so  $|M_2| \leq F \iff |(p_{CoP} - p_0) \cdot t_1| \leq e_1$  and  $|M_1| \leq F \iff |(p_{CoP} - p_0) \cdot t_2| \leq e_2$ : the gauged constraints  $|M_{\{t1\}}| \leq F_n$ ,  $|M_{\{t2\}}| \leq F_n$  say exactly that the centre of pressure lies inside the patch rectangle. In physical units the same statement reads  $|m \cdot t_1| \leq e_2 F_n$ ,  $|m \cdot t_2| \leq e_1 F_n$ , the gauge scalars  $e_2$ ,  $e_1$  being the two half-extents. ■

(Measured, both directions: every nonnegative corner load lands in  $K$  at distance  $0.0$ , and every  $w \in K$  is attained by a nonnegative corner load with nnls residual  $0.0$  [h13 §0, t0.3].)

**Corollary C.10 (uniqueness of the patch solution, and differentiability at a fixed active set).** *Let a patch scene have  $J_{patch}$  of full row rank, so  $G_{patch} > 0$  by Lemma C.7, with  $\lambda_{\min}(G_{patch}) > 0$ . Then for each fixed de Saxcé shift — i.e. at each iterate of the Cadoux outer loop — the cone complementarity problem  $K \ni w \perp G_{patch} w + b^* \in K^*$  is strongly monotone with modulus  $\lambda_{\min}(G_{patch})$  and has a unique solution; no null-space selection rule is required, because  $\ker J_{patch}^T = \{0\}$ . If in addition the active set is locally constant, strict complementarity holds at the solution, and the active-set residual system  $F(w, \mu) = 0$  has a nonsingular Jacobian  $\partial F/\partial w$ , the implicit function theorem gives a differentiable  $w(\mu)$ . Strong monotonicity alone does not give differentiability: at a state where the active set changes, the one-sided directional derivatives of the iteration map differ by 0.511889 [A68 §5].*

*Proof.* Strong monotonicity of the affine map  $w \mapsto G_{patch} w + \hat{b}$  with modulus  $\lambda_{\min}(G_{patch}) > 0$  is  $(w - w')^T (G_{patch} w - G_{patch} w') = (w - w')^T G_{patch} (w - w') \geq \lambda_{\min}(G_{patch}) \|w - w'\|^2$ . A strongly monotone affine variational inequality over a nonempty closed convex cone has exactly one solution (existence by coercivity, uniqueness by subtracting the two variational inequalities). The kernel statement is Lemma C.7. The second statement is the implicit function theorem applied to  $F$  under the stated nonsingularity hypothesis. As one measured example, the Jacobian has  $\text{cond } \partial F/\partial w = 7.259e+01$  at the state  $F_n = 0.5886$ ,  $|M|/F_n = 0.5000$ ,  $\|w_t\|/(\mu F_n) = 1.000000$  [h13 §3C]. That state lies on the boundary of the Contensou cone, so a locally constant active set is an assumption there, not a consequence. ■

This is the algebraic content of §2.5: on the point model  $J_{points}^T$  has a 6-dimensional kernel (Lemma C.6/C.8), the solution set has positive width, and the implicit-function system inherits that singularity ( $\text{cond } \partial F/\partial \lambda = 4.913e+16$ , singular [h13 §3C]); on the patch model the kernel is  $\{0\}$ , the solution is unique, and the same implicit function theorem is applied to a nonsingular system ( $\text{cond } \partial F/\partial w = 7.259e+01$ , analytic  $dw/d\mu$  against central finite differences  $1.886e-10$  [h13 §3C]). The lumping is a change of variables and not a relaxation *on the normal + moment part* — that is Theorem C.9, an equality of cones — while the tangential part of  $K(\mu)$  (the Contensou/ellipsoid coupling of  $(F_{\{t1\}}, F_{\{t2\}}, M_n)$ ) is a genuine model change relative to four independent Coulomb cones, as §2.5 and [h13 §2B] state and as nothing in this appendix claims otherwise.

**What the lumping map conserves, and what the cone change costs.** The map  $C$  conserves virtual work exactly ( $\|J_p^T - J_w^T C\|_\infty \leq 1.3878e-17$ , Lemma C.8 measured), conserves rank exactly (18/18, 48/48, 66/66 rigid-body degrees of freedom, Lemma C.7 measured), and conserves the translation wrench to  $1.83e-10 \dots 5.22e-09$  [U89 §2; A70 §4]. It restores rank and the off-

diagonal sign structure — largest positive off-diagonal eigenvalue  $8.00 \rightarrow 0.0000e+00$ , positive Frobenius share 59–61 %  $\rightarrow 0.00$  % on every stack [U89 §2]. It does not preserve the contact solution: the wrench relative error is 0.3232 on the rotating box and 0.2767 on pure spin [U89 §2; A70 §4]. The appendix therefore states patch lumping as a **rank-restoring approximation**, and Theorem C.9 as the one part of it that is an equality.

The 0.32 is the capacity of the patch cone, not a moment lost by  $C$ .  $C$  has rank 6 and transports the normal torque: tangential corner pairs give  $\tau_n = \sum (x_i f_{\{y,i\}} - y_i f_{\{x,i\}})$ , and  $J_p^\top = J_w^\top C$  carries it [A70 §4]. Four corner Coulomb cones at the rectangle carry  $|\tau_n| = \mu F_n \sqrt{(e_1^2 + e_2^2)}$  under equal normal load and pure spin, while the patch cone  $\|(F_{\{t1\}}, F_{\{t2\}}, \tau_n/\ell)\| \leq \mu F_n$  with  $\ell = \sqrt{((e_1^2 + e_2^2)/3)}$  carries  $\mu F_n \ell$ ; the ratio is  $\sqrt{3} = 1.7320508075688772$ . Measured on the square at  $F_n = 1$ ,  $\mu = 0.5$ : point cone 0.07071067811865475, patch cone 0.040824829046386304 [A70 §4]. A point solution that saturates the corner cones in spin therefore lies outside the patch cone by construction, which is a change of contact law and not a defect of the linear map. Two limits travel with this: the relative errors use the gauged wrench coordinates, so they are not an invariant norm over forces and moments of different units, and the translation test supports equivalence for that test’s symmetry, not for all translation-driven scenes [A70 §4]. The NSD statement belongs to the off-diagonal blocks: the full Delassus operator of the ground-free ring is positive semidefinite ( $\lambda_{\min} = -2.87e-15$ ,  $\lambda_{\max} = 48.3600364878805$ ) while its off-diagonal blocks have largest positive eigenvalue exactly 0.0 [A70 §3].

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## C.6 What the proof does and does not cover

- Lemma C.1 and Corollary C.1.1 are unconditional: the graph assembly  $G = \sum_k (s_k s_k^\top) \otimes w_k$  holds for every scene, degree  $\geq 3$  included. Theorem C.2 is the *sign* statement, and it is exactly there that (C1) and the degree bound (C6) enter.
- Theorem C.3 is unconditional in  $J$ : the spectral corollary of §2.4 does not inherit any of (C1)–(C6). This is the separation §6.3 rests on — the lattice loses the Laplacian, keeps the spectrum.
- Proposition C.5 needs isotropic inverse inertia (point masses); the block statement that replaces it for rigid bodies is Theorem C.2 via Corollary C.2.3.
- Theorem C.9 is the frictionless normal+moment statement for one rectangular 4-corner patch. It says nothing about the frictional part of the patch cone, about non-rectangular patches, or about more than four points; and Corollary C.10’s uniqueness needs full row rank of the whole  $J_{\text{patch}}$ , which fails on closed lattices (rank 48 of 96 [h13 §1H]).
- For any planar polygon patch with non-collinear vertices, the normal-wrench image is  $\{F \geq 0, \text{CoP} \in \text{conv}(\text{vertices})\}$  and the point Jacobian has rank 6. The rectangle’s two box inequalities are replaced by the polygon’s half-plane inequalities; collinear vertices lose a rotational direction [A77 §3].
- No claim in this appendix is measured by this appendix; the numbers cited next to each step were produced by the lanes named; §6.8 records the A77–A82 audit outcomes.

**QED.**

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## References

Bibliographic entries for the external objects named in the text, and for the corpus sources the measurements were taken from. Entries 19, 20, 24, 34 and 35 name objects that appear in the source files rather than in §§1–7 of this paper. This section supersedes the second bullet of Appendix B (which recorded the absence of a bibliography) and, together with Appendix C, the [MISSING] of §2.3; the claims of §§3–7 are unaffected.

Every entry below was checked against the publisher of record in papers/REFERENCES\_P1\_P2.md §P1: of the 36 entries of the previous draft, 27 were verified as written, 8 were corrected and 1 could

not be found. The [VERIFY] marks of the previous draft are removed because that check has been run. Fields for which no primary source could be retrieved are marked **EJ FUNNEN** and are not reconstructed.

## Contact mechanics and the Delassus operator

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## Multibody algorithms

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### Contact solvers and simulators in the NCP lineage

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### Optimization: ADMM parameters and adaptivity

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### Graphs, spectra and linear algebra

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### **Soil mechanics, granular and MPM material models**

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### **Legged-robot stability**

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### **Added in this draft for §6.9**

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